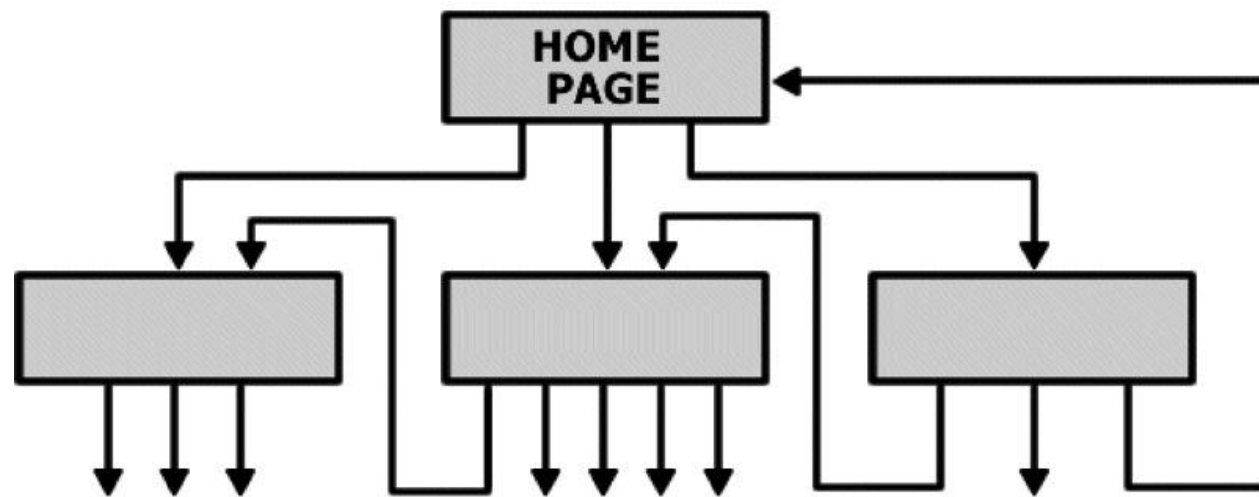


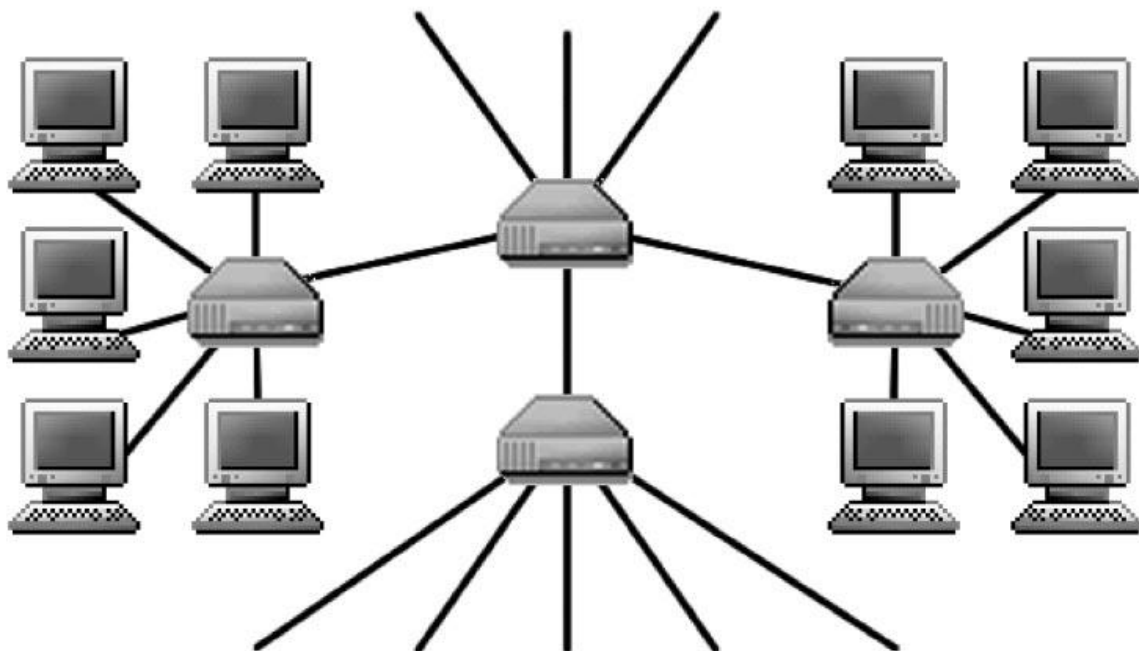
Degree Distribution



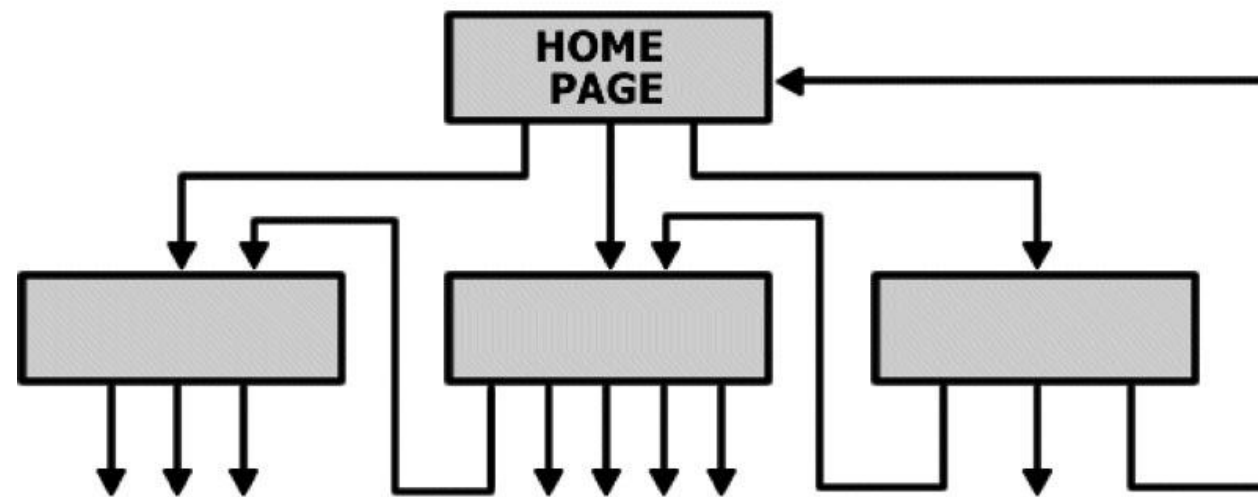
WORLD-WIDE WEB



INTERNET



WORLD-WIDE WEB



INTERNET

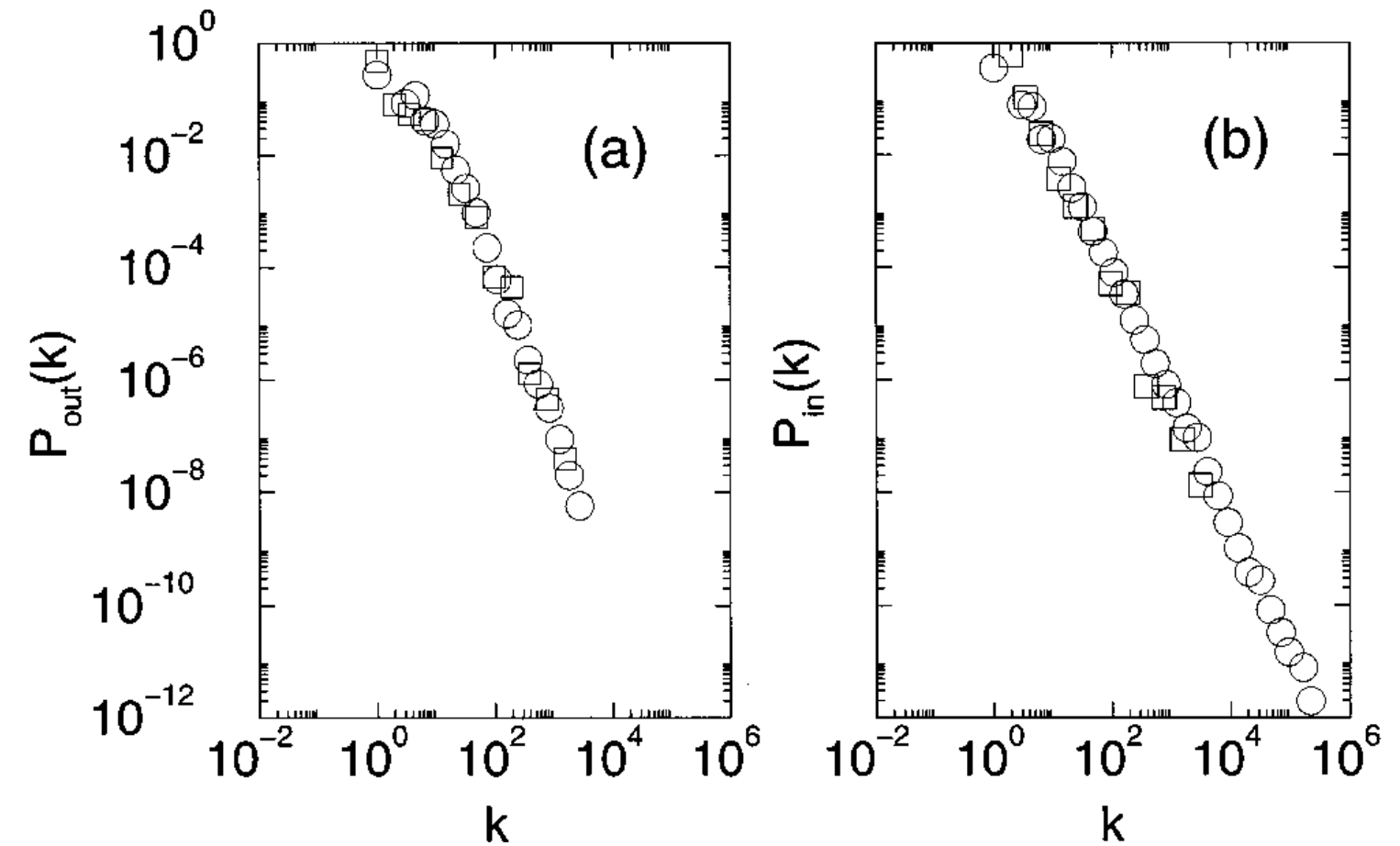
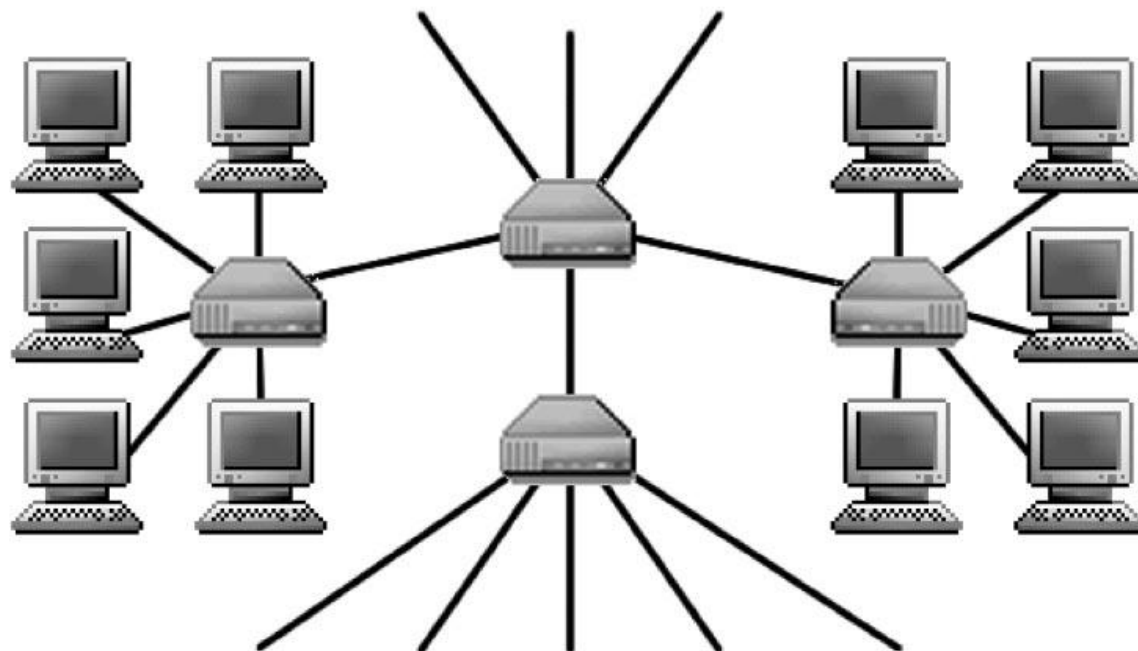
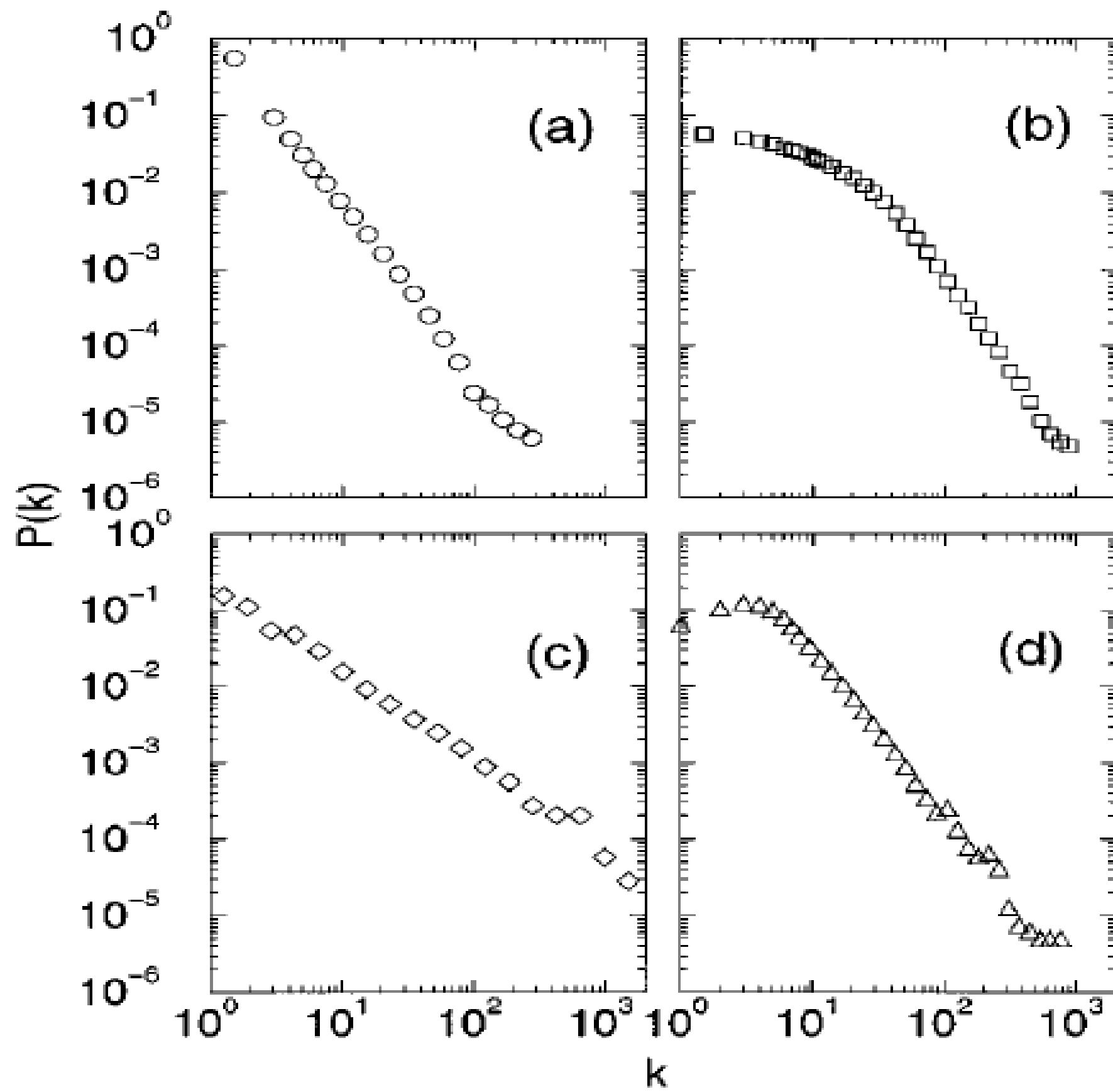


FIG. 2. Degree distribution of the World Wide Web from two different measurements: $\square$, the 325 729-node sample of Albert *et al.* (1999); $\circ$, the measurements of over 200 million pages by Broder *et al.* (2000); (a) degree distribution of the outgoing edges; (b) degree distribution of the incoming edges. The data have been binned logarithmically to reduce noise. Courtesy of Altavista and Andrew Tomkins. The authors wish to thank Luis Amaral for correcting a mistake in a previous version of this figure (see Mossa *et al.*, 2001).



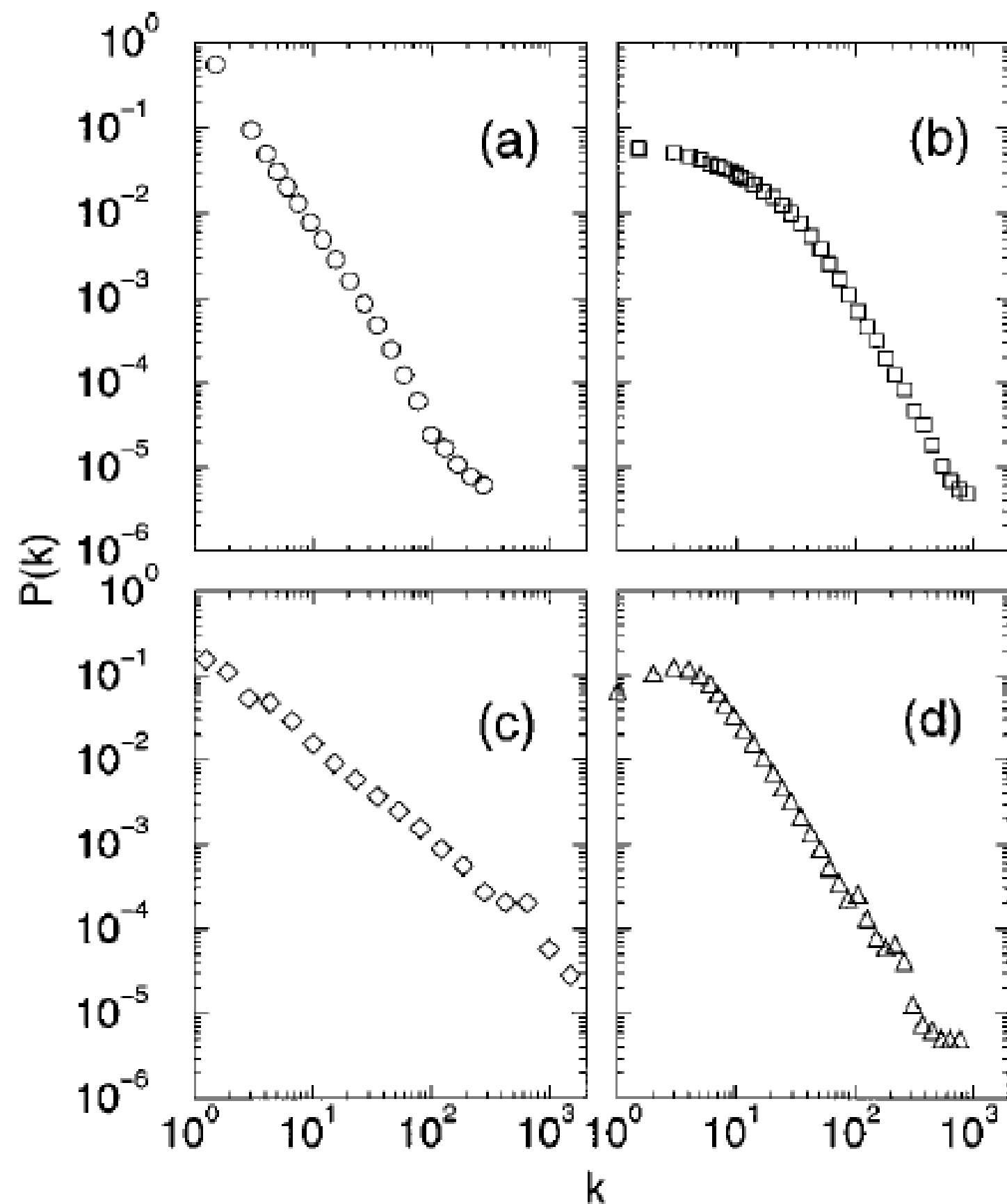


FIG. 3. The degree distribution of several real networks: (a) Internet at the router level. Data courtesy of Ramesh Govindan; (b) movie actor collaboration network. After Barabási and Albert 1999. Note that if TV series are included as well, which aggregate a large number of actors, an exponential cut-off emerges for large k (Amaral *et al.*, 2000); (c) co-authorship network of high-energy physicists. After Newman (2001a, 2001b); (d) co-authorship network of neuroscientists. After Barabási *et al.* (2001).

RMP-Barabasi

Power laws, Pareto distributions and Zipf's law

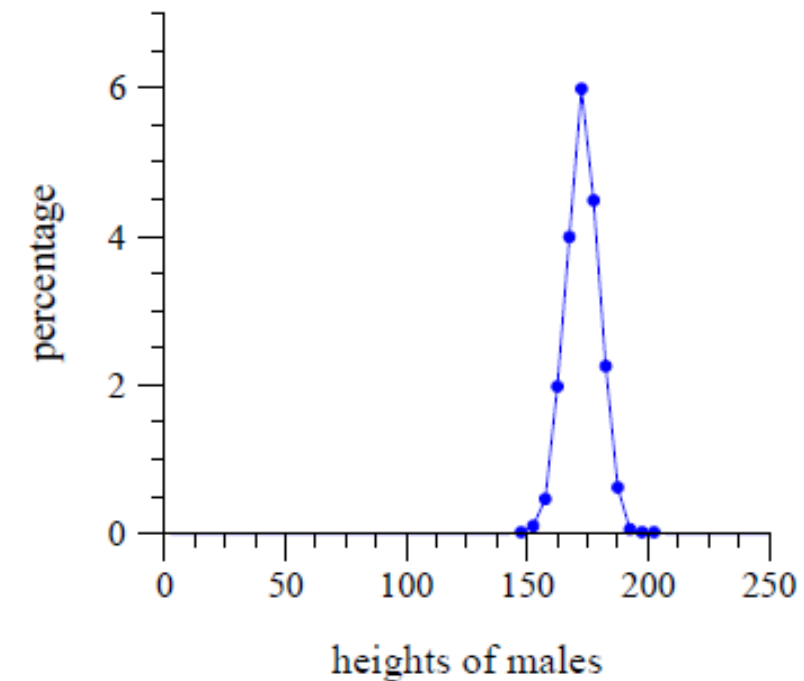
M. E. J. Newman

Department of Physics and Center for the Study of Complex Systems, University of Michigan, Ann Arbor, MI 48109. U.S.A.

- Many of the things that scientists measure **have a typical size or “scale”—a typical value** around which individual measurements are centred.

- Many of the things that scientists measure have a typical size or “scale”—a typical value around which individual measurements are centred.

- **Histogram** of people’s heights



Heights in centimetres of American males. Data from the National Health Examination Survey, 1959–1962 (US Department of Health and Human Services).

<https://aaronclauset.github.io/powerlaws/data.htm>

Histogram : Linear binning

Key Steps in Linear Binning

- 1. Define the Range:** Determine the minimum and maximum values of the dataset.
- 2. Divide into Bins:** Split the range into equal-sized intervals or bins.
- 3. Assign Data Points:** Place each data point into the corresponding bin based on its value.
- 4. Aggregate:** Optionally, compute aggregate statistics (e.g., count, sum, or average) for each bin

Histogram : Linear binning

Key Steps in Linear Binning

- 1. Define the Range:** Determine the minimum and maximum values of the dataset.
- 2. Divide into Bins:** Split the range into equal-sized intervals or bins.
- 3. Assign Data Points:** Place each data point into the corresponding bin based on its value.
- 4. Aggregate:** Optionally, compute aggregate statistics (e.g., count, sum, or average) for each bin

Suppose you have a dataset of values:

[2, 5, 7.3, 13, 15]

- **Range:** Minimum = 2, Maximum = 15
- **Number of Bins:** 3
- **Bin Width:** $(15-2)/3=4.33$ (round to 5 for simplicity)

Histogram : Linear binning

Key Steps in Linear Binning

- 1. Define the Range:** Determine the minimum and maximum values of the dataset.
- 2. Divide into Bins:** Split the range into equal-sized intervals or bins.
- 3. Assign Data Points:** Place each data point into the corresponding bin based on its value.
- 4. Aggregate:** Optionally, compute aggregate statistics (e.g., count, sum, or average) for each bin

Suppose you have a dataset of values:

[2, 5, 7.3, 13, 15]

- **Range:** Minimum = 2, Maximum = 15
- **Number of Bins:** 3
- **Bin Width:** $(15-2)/3=4.33$ (round to 5 for simplicity)

Data points in bins:

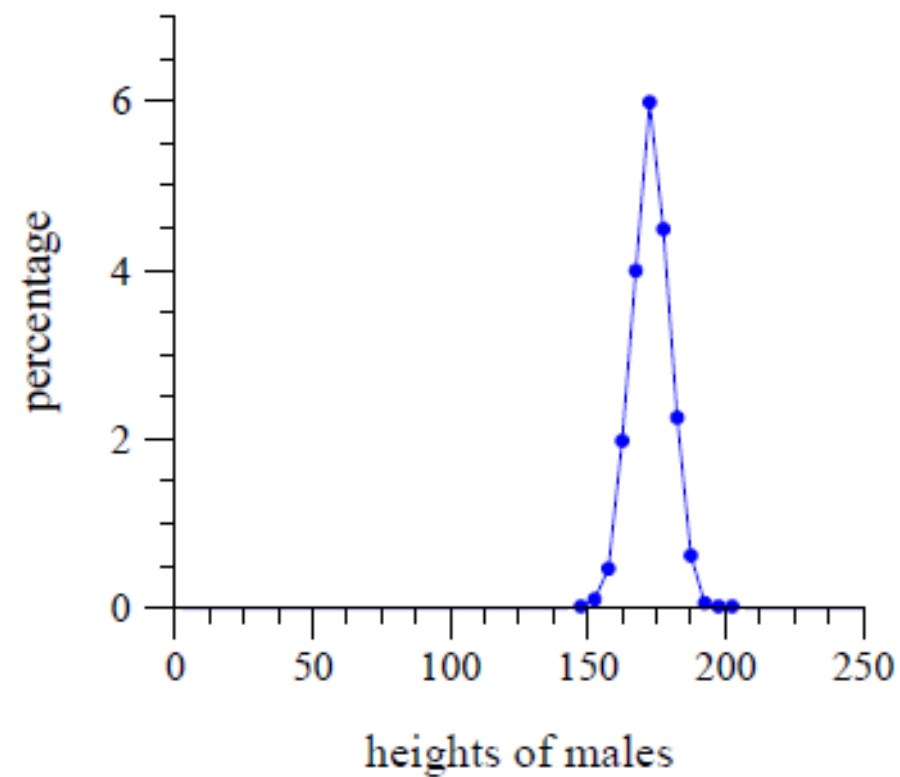
- Bin 1: [2, 5]
- Bin 2: [7.3]
- Bin 3: [13, 15]

Histogram : Linear binning

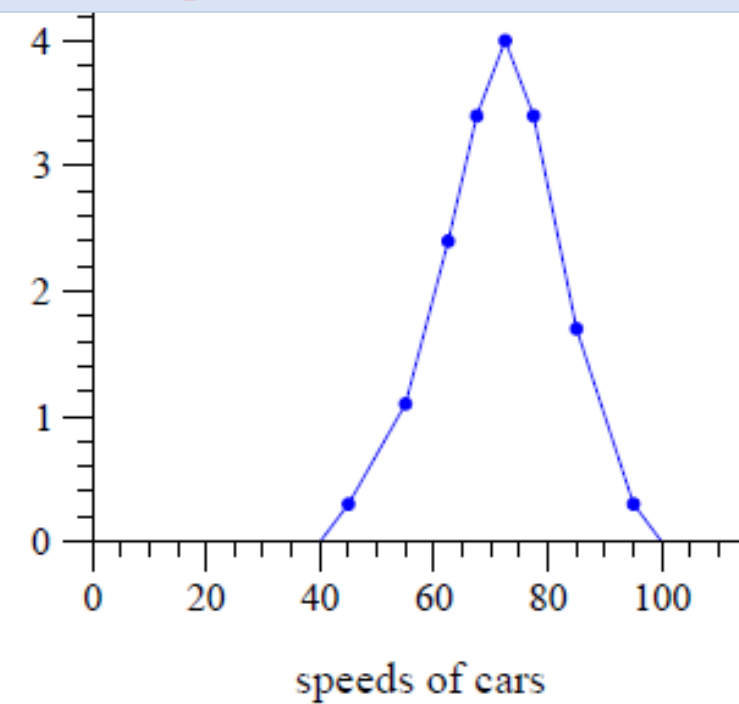
Applications

- Histograms:** To visualize the distribution of data.
- Data Compression:** Reducing the number of data points by grouping them into bins.
- Signal Processing:** Discretizing continuous signals.
- Numerical Methods:** Approximating functions or integrals.

- Many of the things that scientists measure have a typical size or “scale”—a typical value around which individual measurements are centred.



Again the histogram of speeds is strongly peaked, in this case around 75mph.



Right: histogram of speeds in miles per hour of cars on UK motorways. Data from Transport Statistics 2003 (UK Department for Transport).

Power laws, Pareto distributions and Zipf's law

M. E. J. Newman

Department of Physics and Center for the Study of Complex Systems, University of Michigan, Ann Arbor, MI 48109. U.S.A.

But **not all things** we measure **are peaked around a typical value**. Some vary over an enormous dynamic range, sometimes **many orders of magnitude!!!!**

- **frequency table** of words in a text.

$$\text{word frequency} \propto \frac{1}{\text{word rank}} .$$

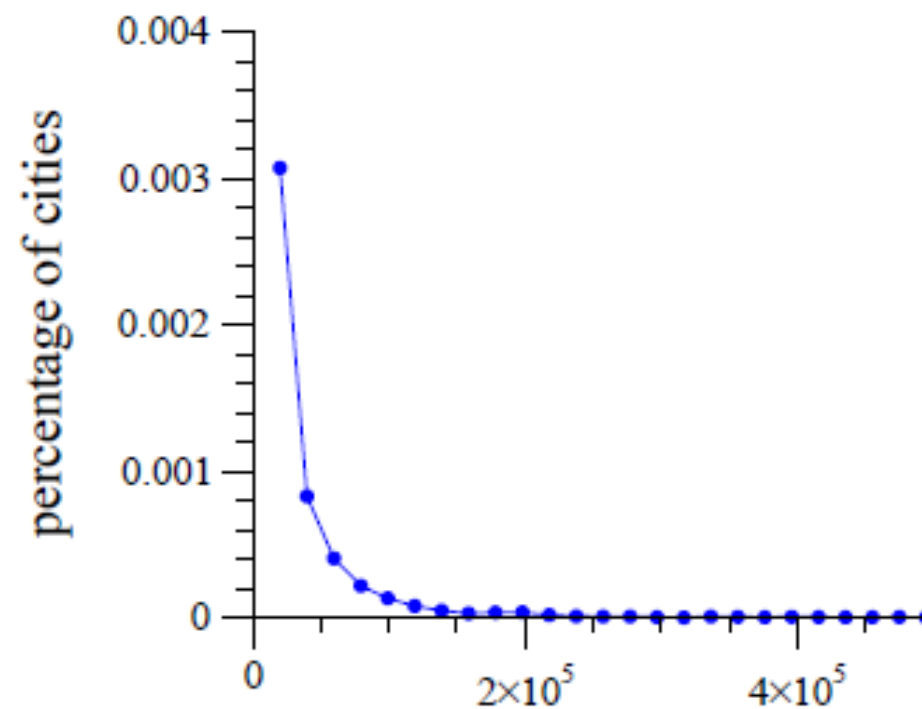


Zipf's law ([/zɪf/](#); German pronunciation: [\[tsɪpʃ\]](#)) is an [empirical law](#) stating that when a list of measured values is sorted in decreasing order, the value of the n -th entry is often approximately [inversely proportional](#) to n .

George Kingsley Zipf ([Statistics](#), [linguistics](#))

But **not all things** we measure **are peaked around a typical value**. Some vary over an enormous dynamic range, sometimes **many orders of magnitude!!!!**

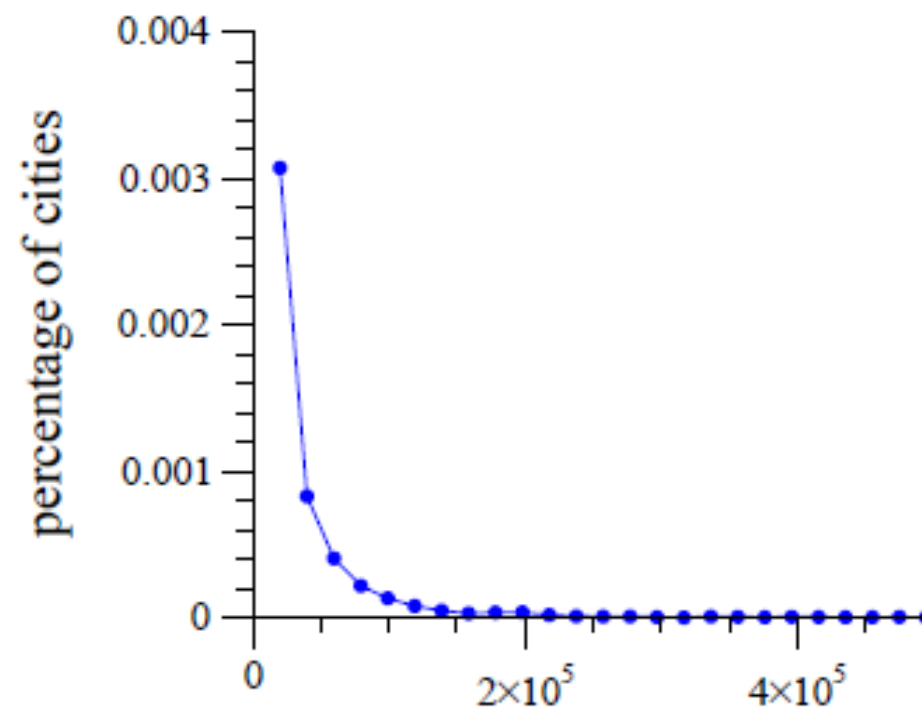
But **not all things** we measure **are peaked around a typical value**. Some vary over an enormous dynamic range, sometimes **many orders of magnitude!!!!**



Histogram of the populations of all US cities with population of 10000 or more.

The largest city in the US is New York City, with over 8.5 million residents. Los Angeles and Chicago follow, each with more than 2.5 million residents, and southern US cities Houston and Phoenix round out the top five with populations of almost 2.3 million and 1.6 million, respectively.

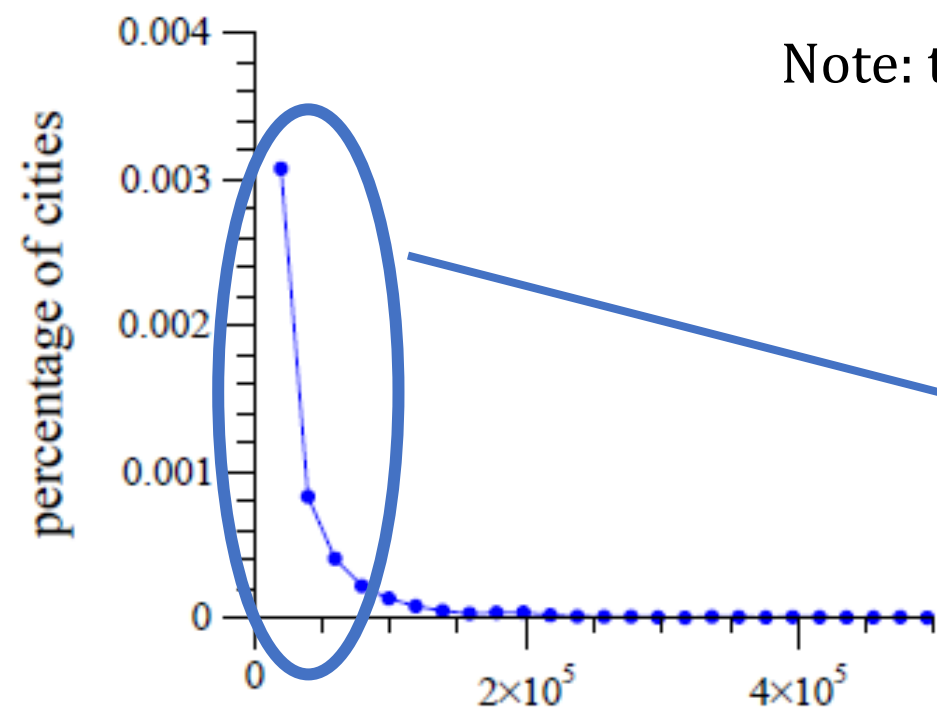
But **not all things** we measure **are peaked around a typical value**. Some vary over an enormous dynamic range, sometimes **many orders of magnitude!!!!**



Histogram of the populations of all US cities with population of 10000 or more.

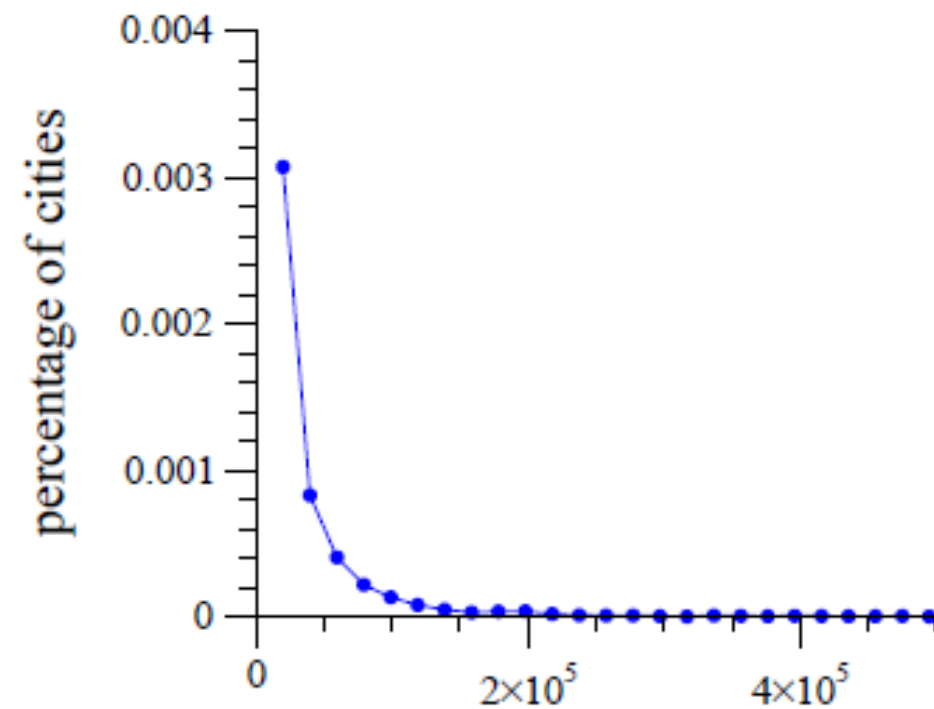
Monowi, Nebraska During the 2000 census, **the village had a total population of two; only one married couple, Rudy and Elsie Eiler, lived there. Rudy died in 2004, leaving his wife as the only resident in the village.** In this capacity, she acts as mayor, and granted herself a liquor license. She is required to produce a municipal road plan every year in order to secure state funding for the village's four street lights.

But **not all things** we measure **are peaked around a typical value**. Some vary over an enormous dynamic range, sometimes **many orders of magnitude!!!!**

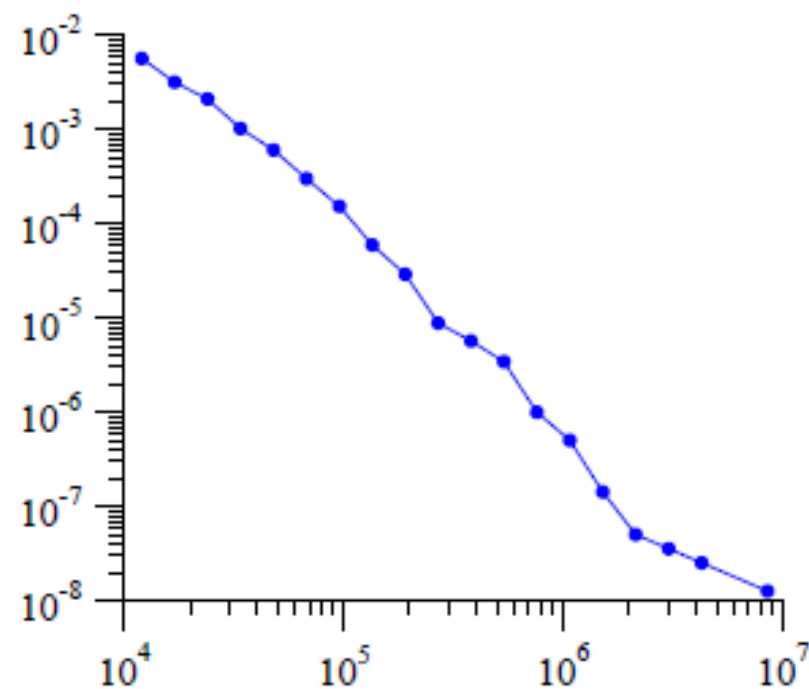


Note: the ratio of largest to smallest population is at least $\sim 10^5$.

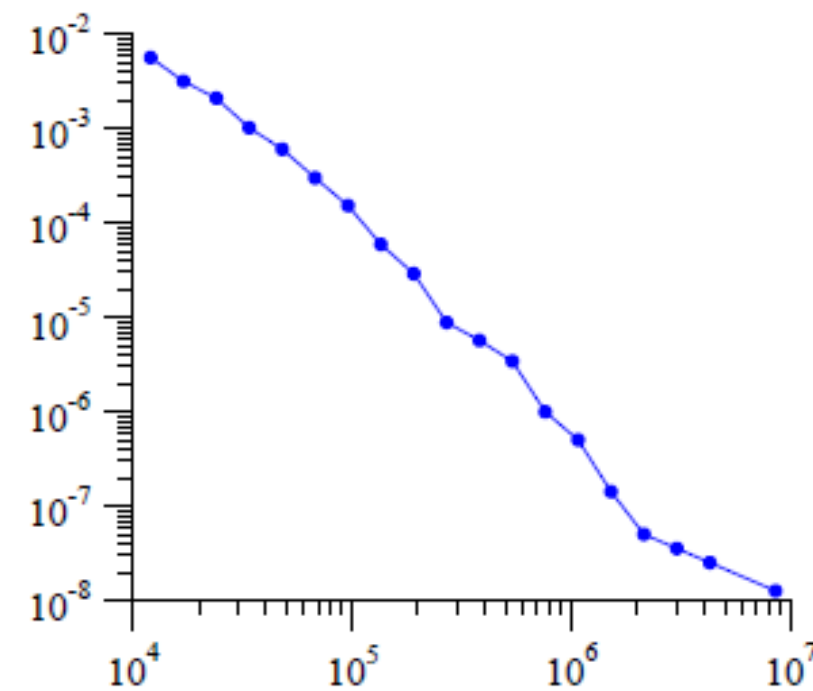
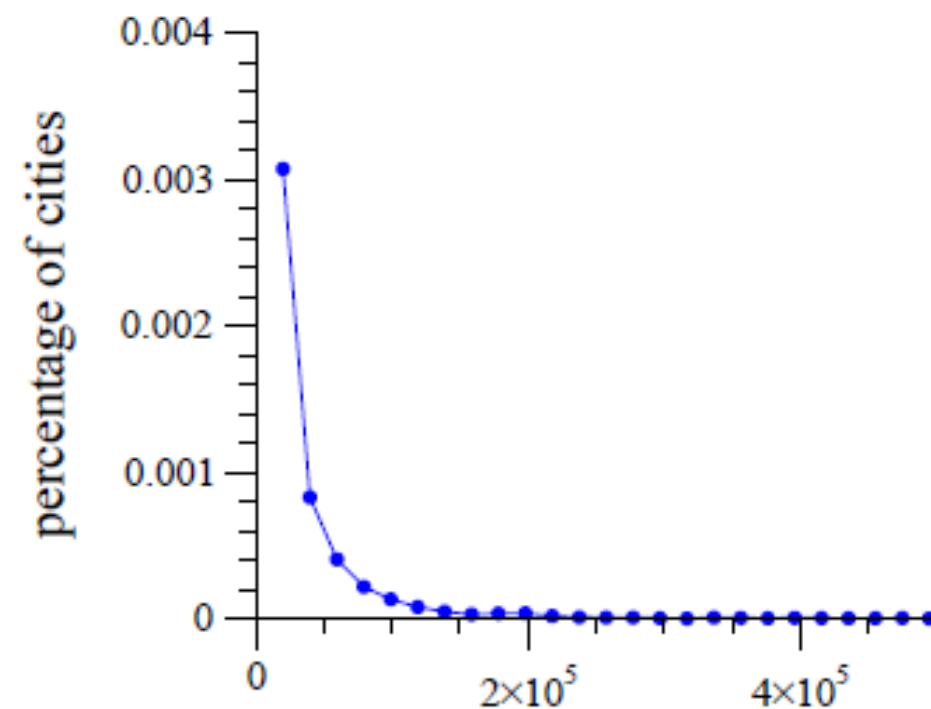
The histogram is **highly right-skewed**, meaning that while the **bulk of the distribution occurs for fairly small sizes** most US cities have small populations—there is a small number of cities with population much higher than the typical value, producing the **long tail to the right** of the histogram.



This right-skewed form is qualitatively quite different from the histograms of people's heights, but is not itself very surprising. Given that we know there is a large dynamic range from the smallest to the largest city sizes, we can immediately deduce that there can only be a small number of very large cities



The histogram of city sizes again, but this time replotted with **logarithmic horizontal and vertical axes**. Now a remarkable pattern emerges: the histogram, when plotted in this fashion, **follows quite closely a straight line**.

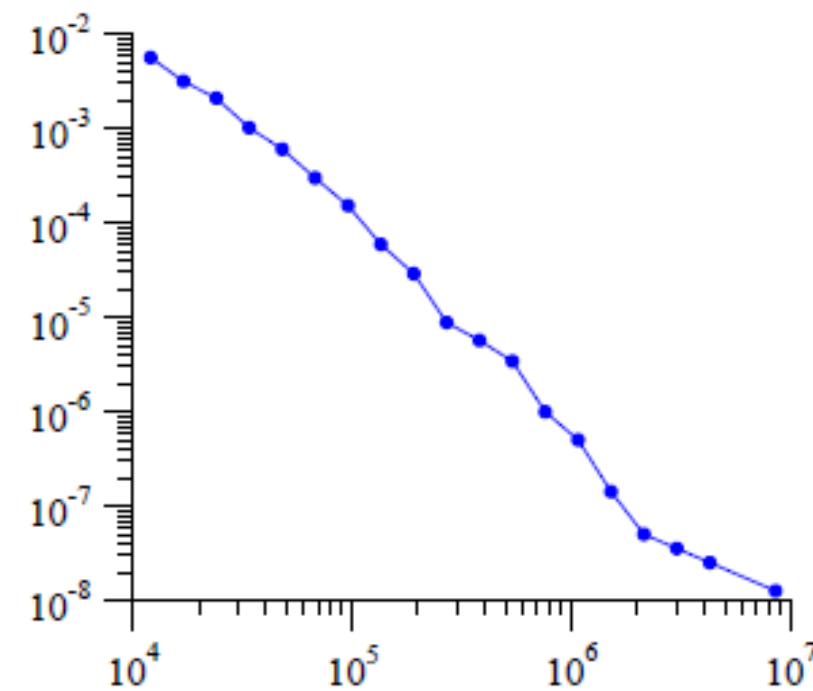
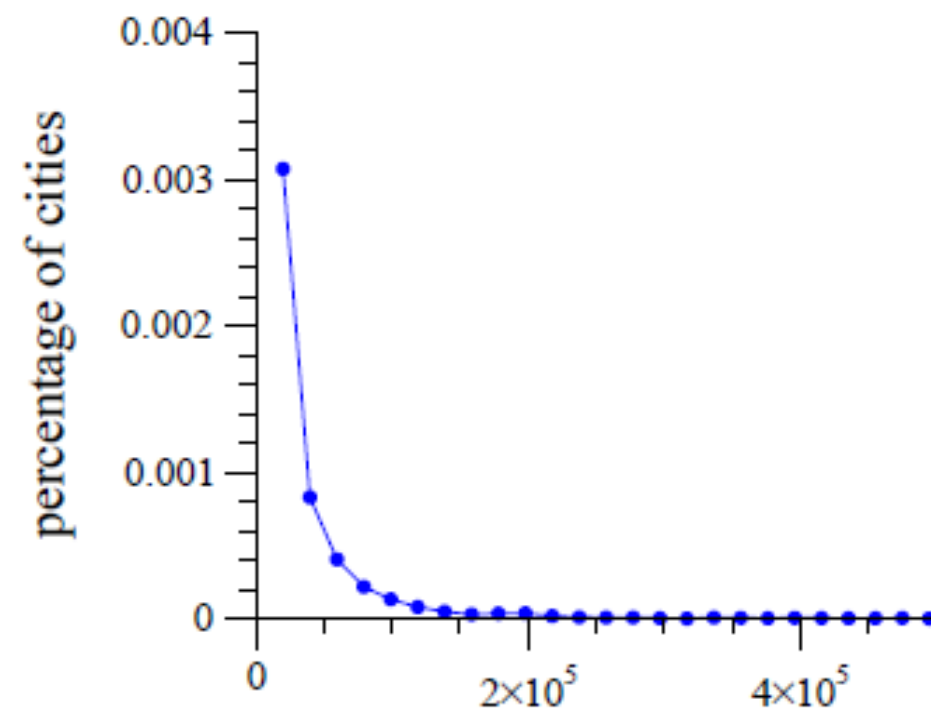


This observation seems first to have been made by Auerbach, although it is often attributed to [Zipf](#). What does it mean?

Let $p(x)dx$ be the fraction of cities with population between x and $x + dx$.

If the histogram is a straight line on log-log scale, then $\ln p(x) = -\alpha \ln(x) + c$, where α and c are constants.

- The slope is negative here.
- This is actually $p(x) = C x^{-\alpha}$, where $c = e^c$,
- This follows a power law.
- (sometimes called as scale-free). Why?

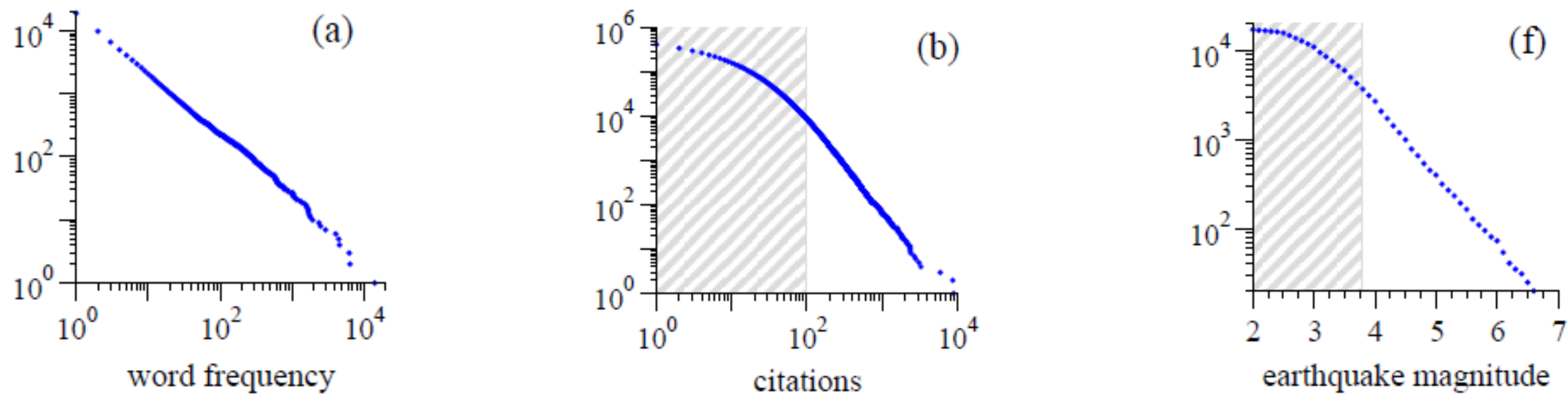


This observation seems first to have been made by Auerbach, although it is often attributed to [Zipf](#). What does it mean?

Let $p(x)dx$ be the fraction of cities with population between x and $x + dx$.

If the histogram is a straight line on log-log scale, then $\ln p(x) = -\alpha \ln(x) + c$, where α and c are constants.

- The slope is negative here.
- This is actually $p(x) = C x^{-\alpha}$, where $c = e^c$,
- This follows a power law.
- (sometimes called as scale-free). Why?



Cumulative distributions or “rank/frequency plots”

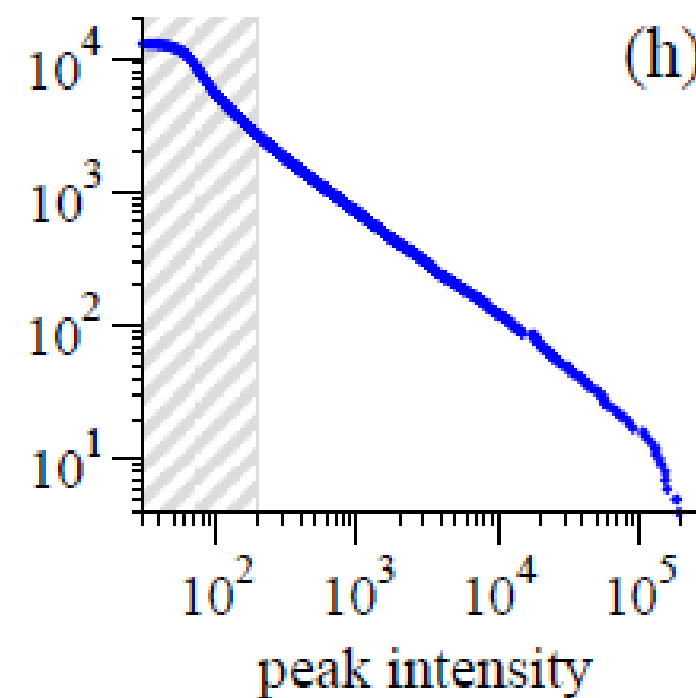
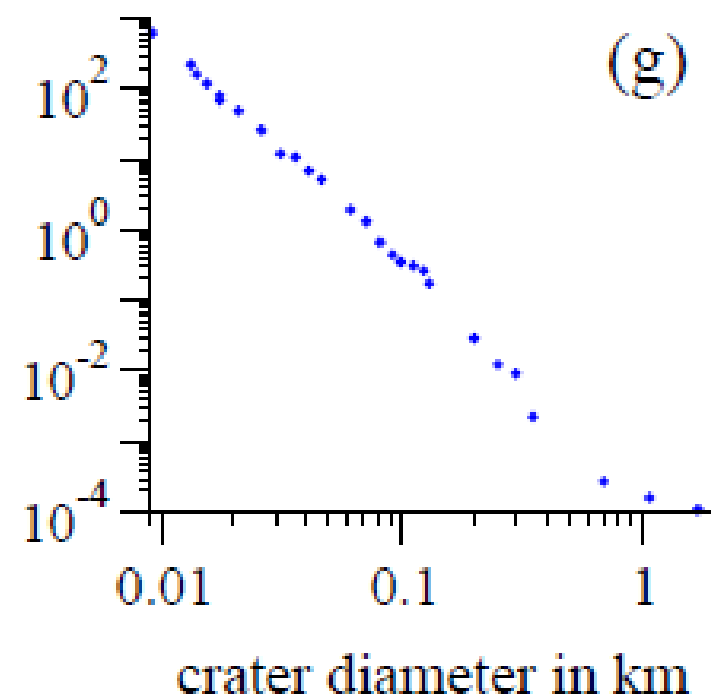
(a) Numbers of occurrences of unique words in the novel Moby Dick by Hermann Melville.

(b) Numbers of citations to scientific papers published in 1981, from time of publication until June 1997.

(f) Magnitude of earthquakes in California between January 1910 and May 1992. The Richter magnitude is defined as the logarithm, base 10, of the maximum amplitude of motion detected in the earthquake, and hence the horizontal scale in the plot, which is drawn as linear, is in effect a logarithmic scale of amplitude. The power law relationship in the earthquake distribution is thus a relationship between amplitude and frequency of occurrence. The data are from the National Geophysical Data Center, www.ngdc.noaa.gov.

<https://aaronclauset.github.io/powerlaws/data.html>

<https://physics.bu.edu/~redner/projects/citation/isi.html>



quantity	minimum	exponent
	$x_{\min}$	α
(a) frequency of use of words	1	2.20(1)
(b) number of citations to papers	100	3.04(2)
(c) number of hits on web sites	1	2.40(1)
(d) copies of books sold in the US	2 000 000	3.51(16)
(e) telephone calls received	10	2.22(1)
(f) magnitude of earthquakes	3.8	3.04(4)
(g) diameter of moon craters	0.01	3.14(5)
(h) intensity of solar flares	200	1.83(2)
(i) intensity of wars	3	1.80(9)
(j) net worth of Americans	\$600m	2.09(4)
(k) frequency of family names	10 000	1.94(1)
(l) population of US cities	40 000	2.30(5)

(g) Diameter of craters on the moon. Vertical axis is measured per square kilometre.

(h) Peak gamma-ray intensity of solar flares in counts per second, measured from Earth orbit between February 1980 and November 1989.

The observations were made between 1980 and 1989 by the instrument known as the Hard X-Ray Burst Spectrometer aboard the Solar Maximum Mission satellite launched in 1980. The spectrometer used a CsI scintillation detector to measure gamma-rays from solar flares and the horizontal axis in the figure is calibrated in terms of scintillation counts per second from this detector. The data are from the NASA Goddard Space Flight Center, umbra.nascom.nasa.gov/smm/hxrbs.html.

But these plots are cumulative!!

Power-law distributions occur in an extraordinarily diverse range of phenomena. In addition to city populations,

- The sizes of earthquakes [3],
- Moon craters [4],
- Solar flares [5],
- Computer files [6] and wars [7],
- The frequency of use of words in any human language [2, 8],
- The frequency of occurrence of personal names in most cultures [9], the numbers of papers scientists write [10], the number of citations received by papers [11],
- The number of hits on web pages [12],
- The sales of books, music recordings and almost every other branded commodity [13, 14],
- The numbers of species in biological taxa [15],
- People's annual incomes [16] follows power law distributions.

[3] B. Gutenberg and R. F. Richter, Frequency of earthquakes in California. *Bulletin of the Seismological Society of America* 34, 185–188 (1944).

[4] G. Neukum and B. A. Ivanov, Crater size distributions and impact probabilities on Earth from lunar, terrestrial planet, and asteroid cratering data. In T. Gehrels (ed.), *Hazards Due to Comets and Asteroids*, pp. 359–416, University of Arizona Press, Tucson, AZ (1994).

[5] E. T. Lu and R. J. Hamilton, Avalanches of the distribution of solar flares. *Astrophysical Journal* 380, 89–92 (1991).

[6] M. E. Crovella and A. Bestavros, Self-similarity in World Wide Web traffic: Evidence and possible causes. In B. E. Gaither and D. A. Reed (eds.), *Proceedings of the 1996 ACM SIGMETRICS Conference on Measurement and Modeling of Computer Systems*, pp. 148–159, Association of Computing Machinery, New York (1996).

[7] D. C. Roberts and D. L. Turcotte, Fractality and self-organized criticality of wars. *Fractals* 6, 351–357 (1998).

[8] J. B. Estoup, *Gammes Sténographiques*. Institut Sténographique de France, Paris (1916).

[9] D. H. Zanette and S. C. Manrubia, Vertical transmission of culture and the distribution of family names. *Physica A* 295, 1–8 (2001).

[10] A. J. Lotka, The frequency distribution of scientific production. *J. Wash. Acad. Sci.* 16, 317–323 (1926).

[11] D. J. de S. Price, Networks of scientific papers. *Science* 149, 510–515 (1965).

[12] L. A. Adamic and B. A. Huberman, The nature of markets in the World Wide Web. *Quarterly Journal of Electronic Commerce* 1, 512 (2000).

[13] R. A. K. Cox, J. M. Felton, and K. C. Chung, The concentration of commercial success in popular music: an analysis of the distribution of gold records. *Journal of Cultural Economics* 19, 333–340 (1995).

[14] R. Kohli and R. Sah, Market shares: Some power law results and observations. Working paper 04.01, Harris School of Public Policy, University of Chicago (2003).

[15] J. C. Willis and G. U. Yule, Some statistics of evolution and geographical distribution in plants and animals, and their significance. *Nature* 109, 177–179 (1922).

[16] V. Pareto, *Cours d'Economie Politique*. Droz, Geneva (1896).

Histogram : Linear binning

Key Steps in Linear Binning

- 1. Define the Range:** Determine the minimum and maximum values of the dataset.
- 2. Divide into Bins:** Split the range into equal-sized intervals or bins.
- 3. Assign Data Points:** Place each data point into the corresponding bin based on its value.
- 4. Aggregate:** Optionally, compute aggregate statistics (e.g., count, sum, or average) for each bin

Histogram : Linear binning

Key Steps in Linear Binning

- 1. Define the Range:** Determine the minimum and maximum values of the dataset.
- 2. Divide into Bins:** Split the range into equal-sized intervals or bins.
- 3. Assign Data Points:** Place each data point into the corresponding bin based on its value.
- 4. Aggregate:** Optionally, compute aggregate statistics (e.g., count, sum, or average) for each bin

Suppose you have a dataset of values:

[2, 5, 7.3, 13, 15, 16]

- **Range:** Minimum = 2, Maximum = 16
- **Number of Bins:** 3
- **Bin Width:** $(15-2)/3=4.6$ (round to 5 for simplicity)

Histogram : Linear binning

Key Steps in Linear Binning

- 1. Define the Range:** Determine the minimum and maximum values of the dataset.
- 2. Divide into Bins:** Split the range into equal-sized intervals or bins.
- 3. Assign Data Points:** Place each data point into the corresponding bin based on its value.
- 4. Aggregate:** Optionally, compute aggregate statistics (e.g., count, sum, or average) for each bin

Suppose you have a dataset of values:

[2, 5, 7.3, 13, 15, 16]

- **Range:** Minimum = 2, Maximum = 16
- **Number of Bins:** 3
- **Bin Width:** $(15-2)/3=4.6$ (round to 5 for simplicity)

Data points in bins:

- Bin 1: [2, 5]
- Bin 2: [7.3]
- Bin 3: [13, 15, 16]

Histogram : Linear binning

Applications

- Histograms:** To visualize the distribution of data.
- Data Compression:** Reducing the number of data points by grouping them into bins.
- Signal Processing:** Discretizing continuous signals.
- Numerical Methods:** Approximating functions or integrals.

Histogram : Log binning

- Log binning is a technique used in data analysis, particularly for dealing with data that **spans several orders of magnitude**, such as distributions that follow **power laws**.
- Instead of using linear bin widths, the **bins are spaced logarithmically**, allowing better representation of data in cases where the frequency of occurrence decreases exponentially with increasing values.

Histogram : Log binning

Key Features of Log Binning:

1.Logarithmic Bin Widths:

1. The width of the bins *increases logarithmically*, meaning the difference between bin boundaries grows as the values increase.
2. For example, bin edges might be spaced like 1,10, 100, 1000 rather than 1,2,3,4.

2.Preserves Detail Across Scales:

1. Small bins are used for low-value ranges, capturing fine details.
2. Larger bins are used for high-value ranges, ensuring sparsely populated regions are still well-represented.

Histogram : Log binning

Key Features of Log Binning:

1.Logarithmic Bin Widths:

1. The width of the bins *increases logarithmically*, meaning the difference between bin boundaries grows as the values increase.
2. For example, bin edges might be spaced like 1,10, 100, 1000 rather than 1,2,3,4.

2.Preserves Detail Across Scales:

1. Small bins are used for low-value ranges, capturing fine details.
2. Larger bins are used for high-value ranges, ensuring sparsely populated regions are still well-represented.

3.Reduces Noise:

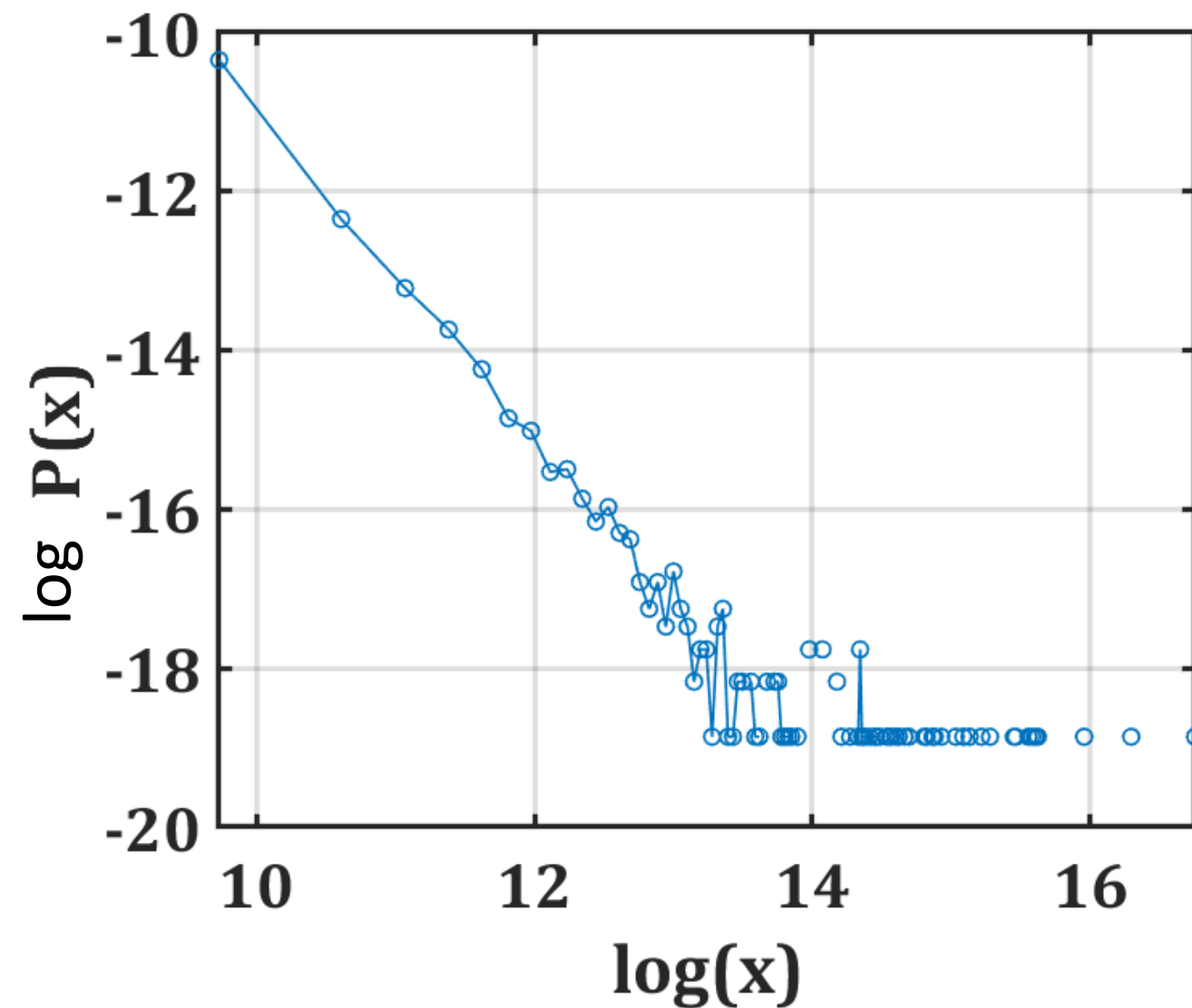
1. In distributions with a long tail (e.g., power-law distributions), linear binning can lead to sparse or empty bins for large values. Log binning aggregates these values more effectively, reducing statistical noise.

4.Logarithmic Representation:

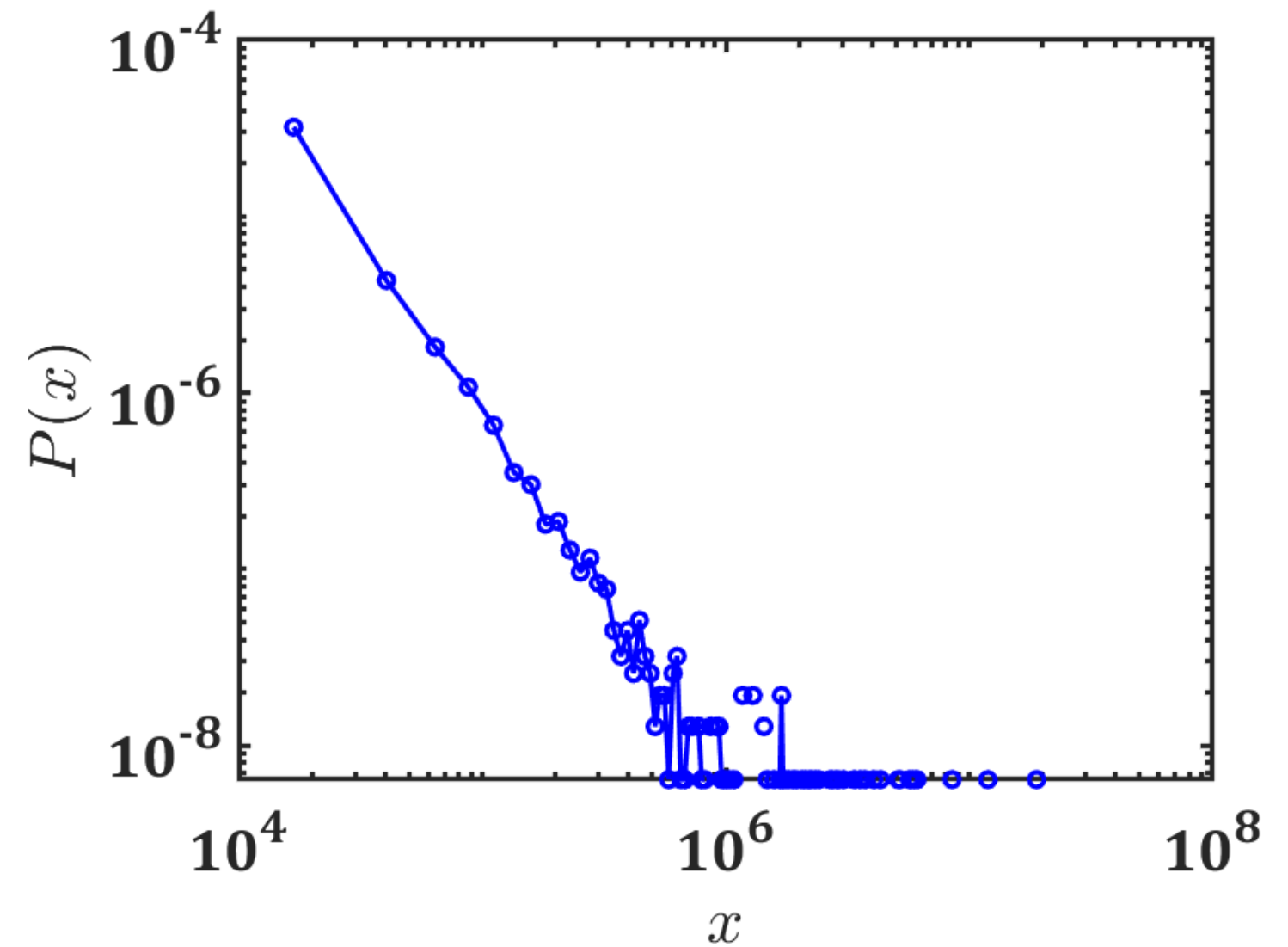
1. Data is often plotted on a log-log scale for visualization. The x-axis represents the bin range, and the y-axis shows the frequency or probability density.

Histogram (US population)

Linear binning and log of the binned data

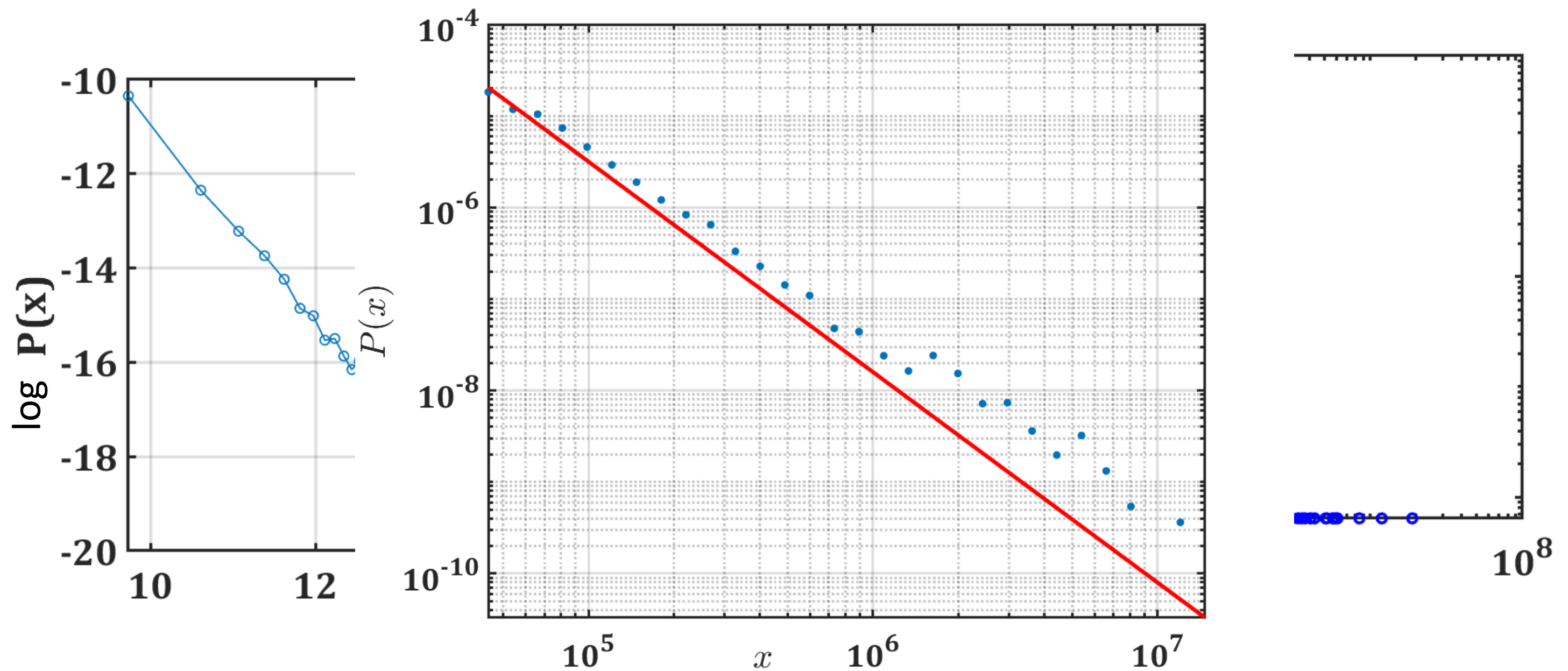


Linear binning and log-scale



Histogram (US population)

Log binning and log scale



Histogram : Log binning

- 1. Choose the Base of the Logarithm:** Decide the base for your logarithmic binning (e.g., base-2, base-10, or natural log). This determines the width of the bins.
- 2. Determine the Range of Data:** Identify the minimum and maximum values of the dataset. These will define the range for your bins.
- 3. Define the Bin Edges:** Create bin edges using logarithmic intervals. For instance, if the base is 10:
Bin edges = $\{10^0, 10^1, 10^2, \dots\}$
Or, for smaller steps:
Bin edges = $\{10^0, 10^{0.1}, 10^{0.2}, \dots\}$
- 4. Assign Data Points to Bins:** For each data point, determine which bin it belongs to by comparing its value to the bin edges. In practice, you can use algorithms for efficient bin allocation.
- 5. Count Data Points in Each Bin:** After assigning all data points to bins, count the number of points in each bin.
- 6. Normalize Bin Counts :** Normalize the bin counts by the bin width to account for the fact that the bin sizes grow logarithmically.
- 7. Plot the Results:** Plot the histogram, often using a log-log scale to reveal power-law behavior or other patterns.

Histogram : Linear binning and Log binning

Task : Histogram of the set of random numbers which will have a power-law distribution with exponent $\alpha = 2.5$.

$$p(x) = \frac{\alpha - 1}{x_{\min}} \left(\frac{x}{x_{\min}} \right)^{-\alpha}.$$

Use transformation method.

Generate a random real number r uniformly distributed in the range $0 \leq r < 1$, then $x = x_{\min}(1 - r)^{-1/(\alpha-1)}$ is a random power-law-distributed real number in the range $x_{\min} \leq x < \infty$ with exponent α .

Note that there has to be a lower limit $x_{\min}$ on the range; the power-law distribution diverges as $x \rightarrow 0$

Histogram : Linear binning vs Log binning

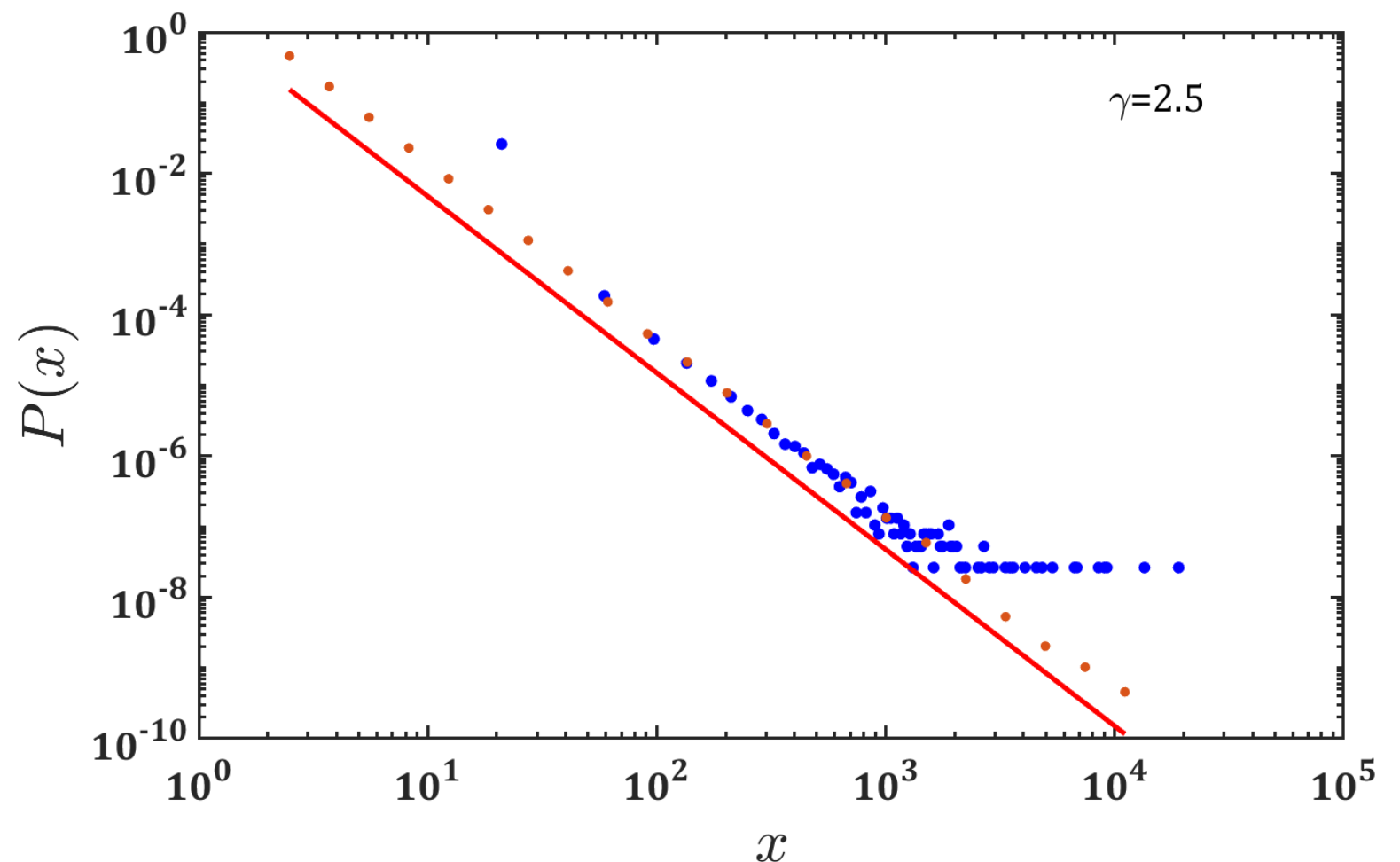
(Synthetic data)

Task : Histogram of the set of random numbers which will have a power-law distribution with exponent $\alpha = 2.5$.

Use transformation method.

Generate a random real number r uniformly distributed in the range $0 \leq r < 1$, then $x = x_{\min}(1 - r)^{-1/(\alpha-1)}$ is a random power-law-distributed real number in the range $x_{\min} \leq x < \infty$ with exponent α .

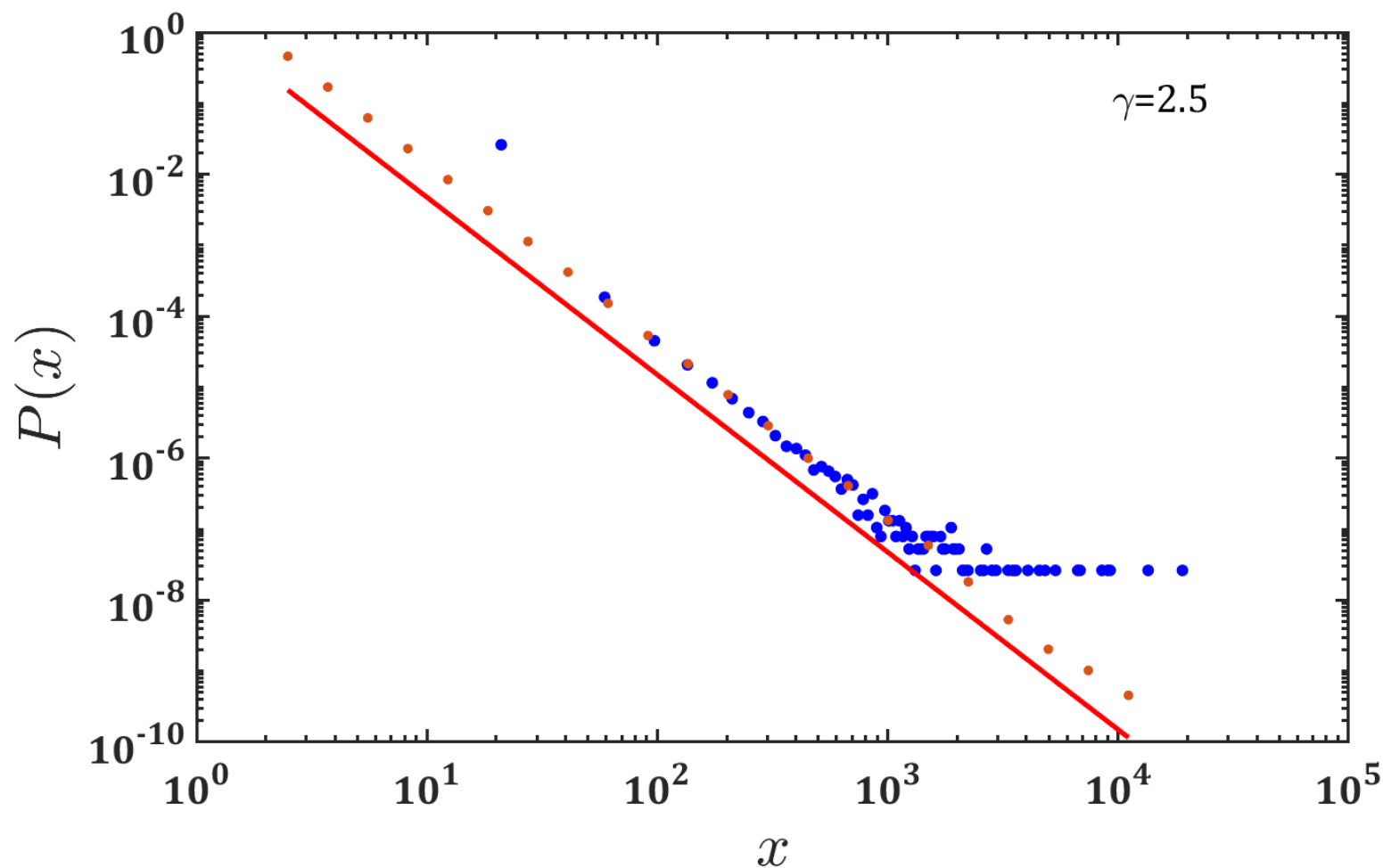
Note that there has to be a lower limit $x_{\min}$ on the range; the power-law distribution diverges as $x \rightarrow 0$



Histogram : Linear binning vs Log binning

(Synthetic data)

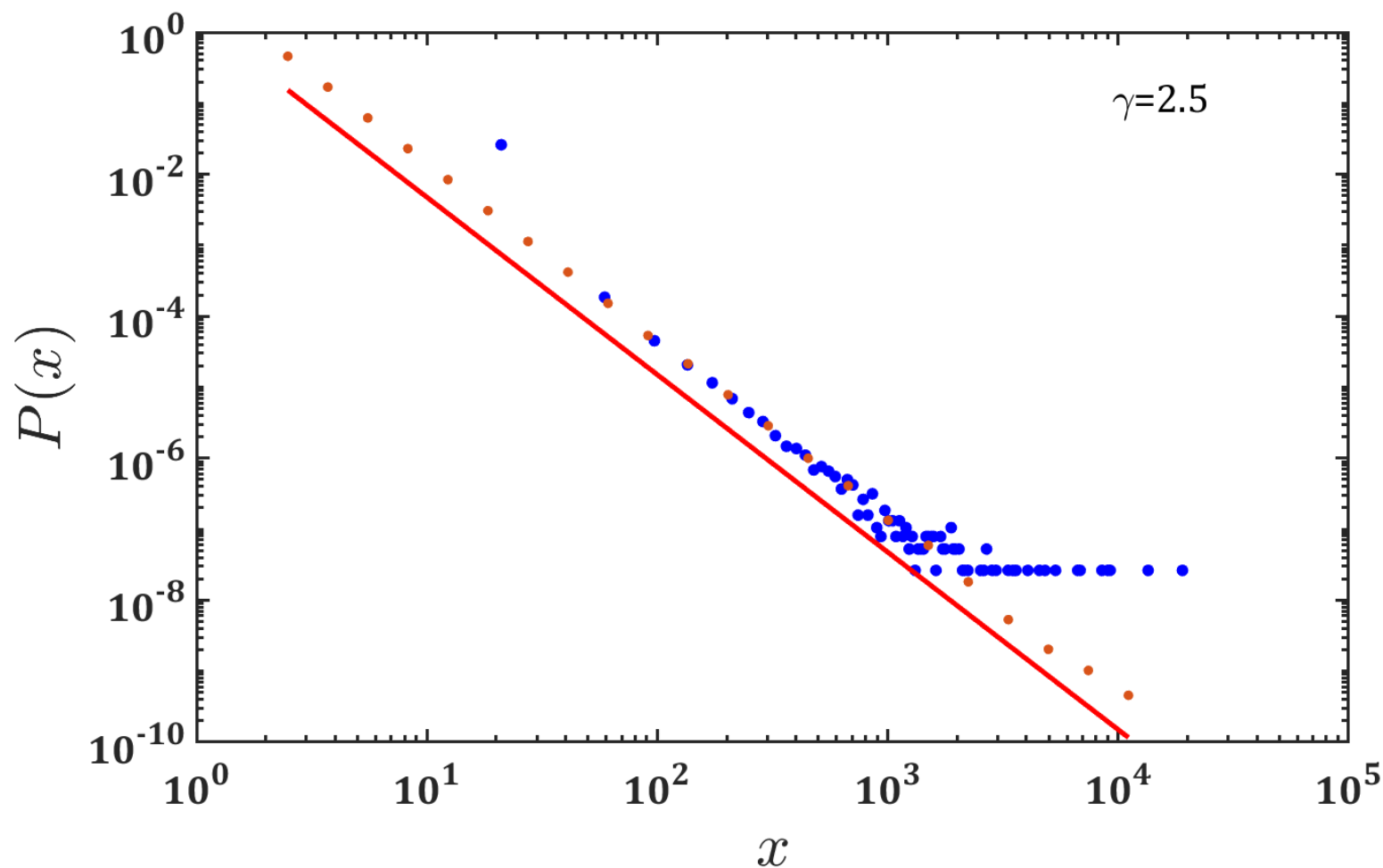
Task : Histogram of the set of random numbers which will have a power-law distribution with exponent $\alpha = 2.5$.



- To reveal the power-law form of the distribution it is better, as we have seen, to plot the histogram on logarithmic scales, and when we do this for the current data we see the characteristic straight-line form of the power-law distribution (blue dots).
- However, the plot is in some respects not a very good one. In particular the right-hand end of the distribution is noisy because of sampling errors.
- The power-law distribution dwindles in this region, meaning that each bin only has a few samples in it, if any. So the fractional fluctuations in the bin counts are large and this appears as a noisy curve on the plot.

Histogram : Linear binning vs Log binning

Task : Histogram of the set of random numbers which will have a power-law distribution with exponent $\alpha = 2.5$.



- **Solution:** Vary the width of the bins in the histogram. If we are going to do this, we must also normalize the sample counts by the width of bins they fall in. That is, the number of samples in a bin of width Δx should be divided by Δx to get a count per unit interval of x .
- Then the normalized sample count becomes independent of bin width on average and we are free to vary the bin widths as we like.

The most common choice is to create bins such that each is a fixed multiple wider than the one before it. This is known as **logarithmic binning**.

A continuous real variable with a power-law distribution has a probability $p(x) \, dx$ of taking a value in the interval from x to $x + dx$, where

$$p(x) = Cx^{-\alpha},$$

with $\alpha > 0$.

The constant C is given by the normalization requirement that

$$1 = \int_{x_{\min}}^{\infty} p(x) \, dx = C \int_{x_{\min}}^{\infty} x^{-\alpha} \, dx = \frac{C}{1-\alpha} \left[x^{-\alpha+1} \right]_{x_{\min}}^{\infty}. \quad C = (\alpha - 1)x_{\min}^{\alpha-1},$$

$$p(x) = \frac{\alpha - 1}{x_{\min}} \left(\frac{x}{x_{\min}} \right)^{-\alpha}.$$

$$p(x) = \frac{\alpha - 1}{x_{\min}} \left(\frac{x}{x_{\min}} \right)^{-\alpha}.$$

The mean value of our power-law distributed quantity x is given by

$$\begin{aligned} \langle x \rangle &= \int_{x_{\min}}^{\infty} xp(x) \, dx = C \int_{x_{\min}}^{\infty} x^{-\alpha+1} \, dx \\ &= \frac{C}{2 - \alpha} \left[x^{-\alpha+2} \right]_{x_{\min}}^{\infty}. \end{aligned}$$

Note that this expression becomes infinite if $\alpha \leq 2$. Power laws with such low values of α have no finite mean. The distributions of sizes of solar flares and wars in are examples of such power laws.

$$p(x) = \frac{\alpha - 1}{x_{\min}} \left(\frac{x}{x_{\min}} \right)^{-\alpha}.$$

For $\alpha > 2$ however, the mean is perfectly well defined, with a value given by

$$\langle x \rangle = \frac{\alpha - 1}{\alpha - 2} x_{\min}.$$

We can also calculate higher moments of the distribution $p(x)$. For instance, the second moment, the mean square, is given by

$$\langle x^2 \rangle = \frac{C}{3 - \alpha} \left[x^{-\alpha+3} \right]_{x_{\min}}^{\infty}.$$

For $\alpha > 2$ however, the mean is perfectly well defined, with a value given by Eq. (11) of

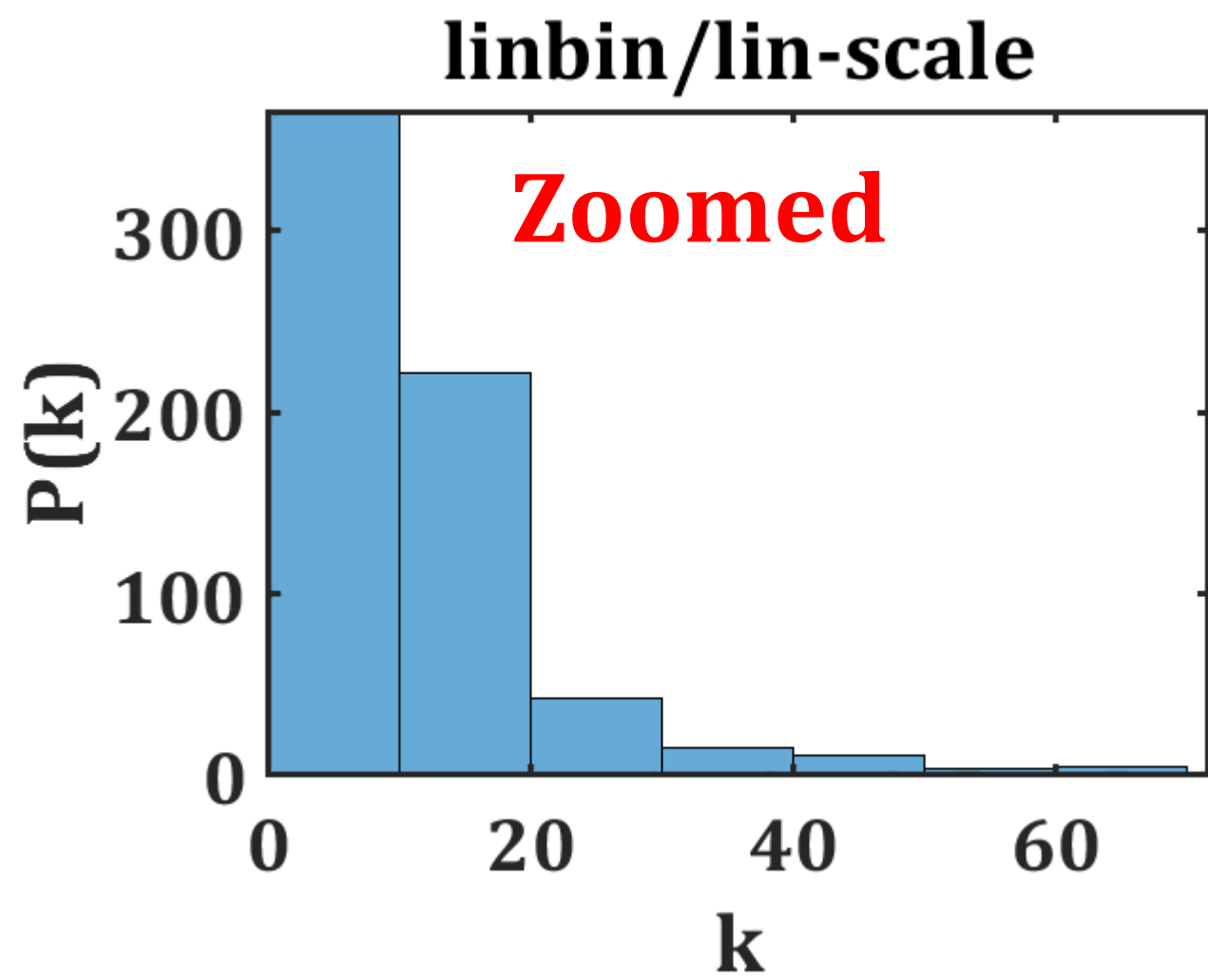
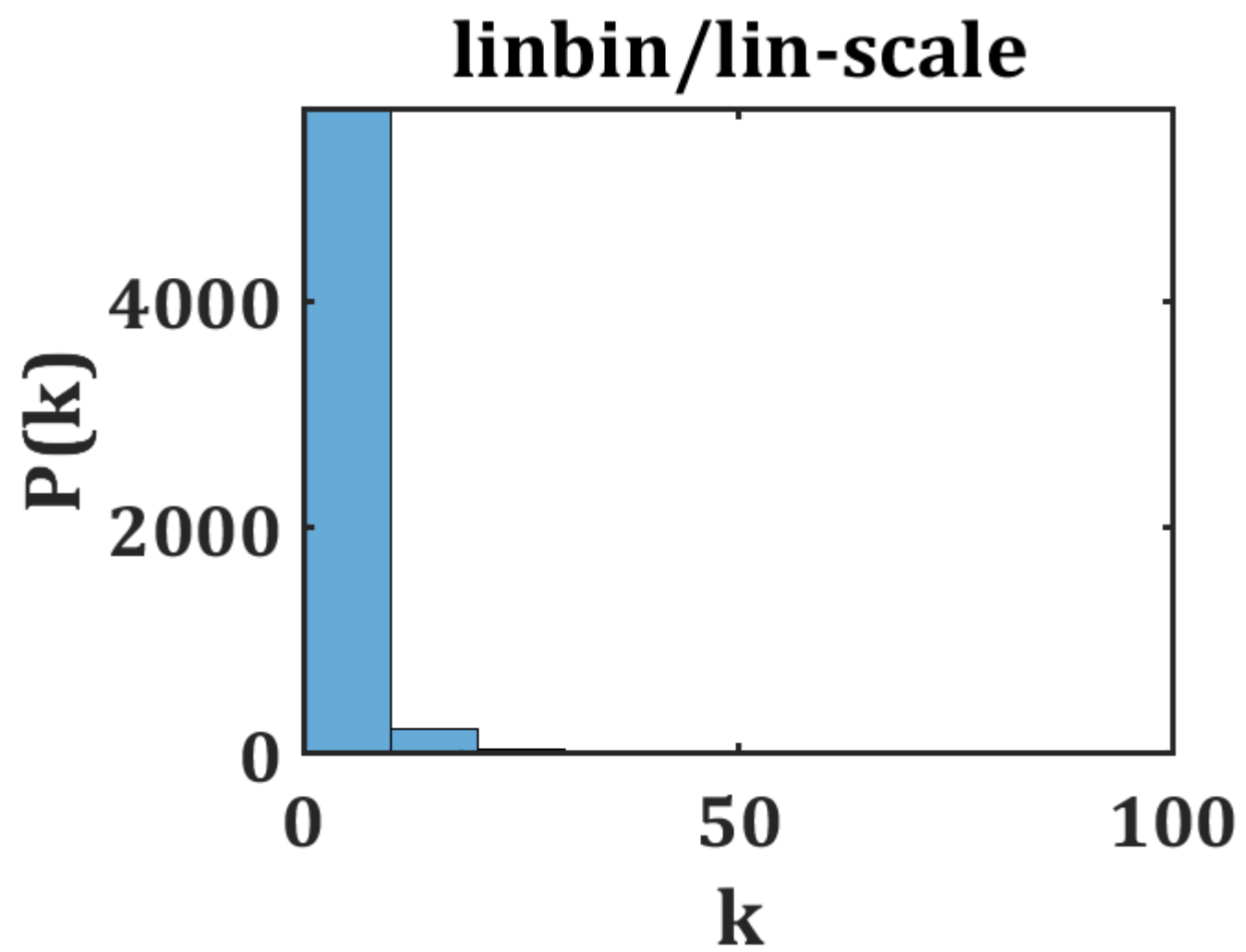
$$\langle x \rangle = \frac{\alpha - 1}{\alpha - 2} x_{\min}.$$

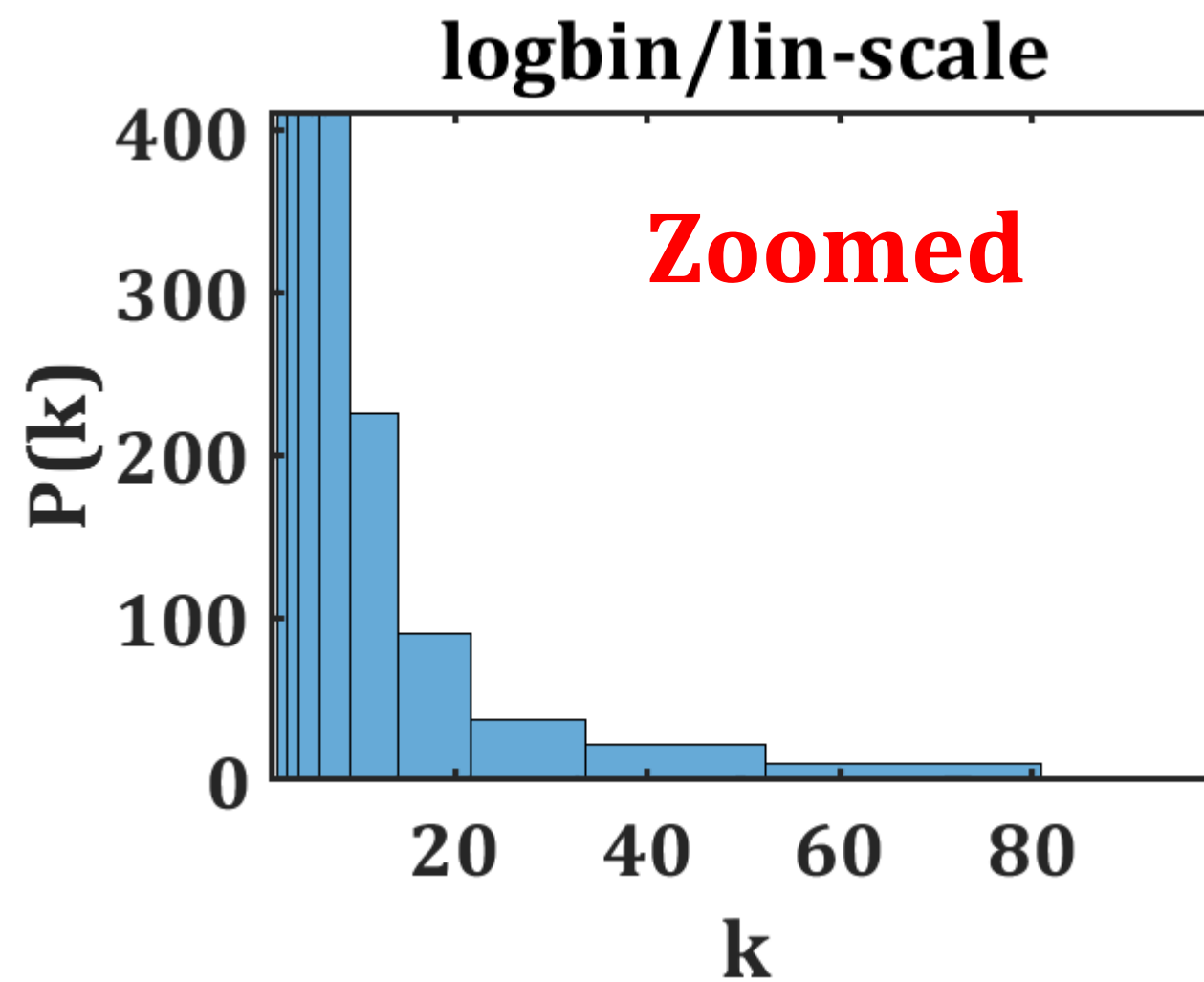
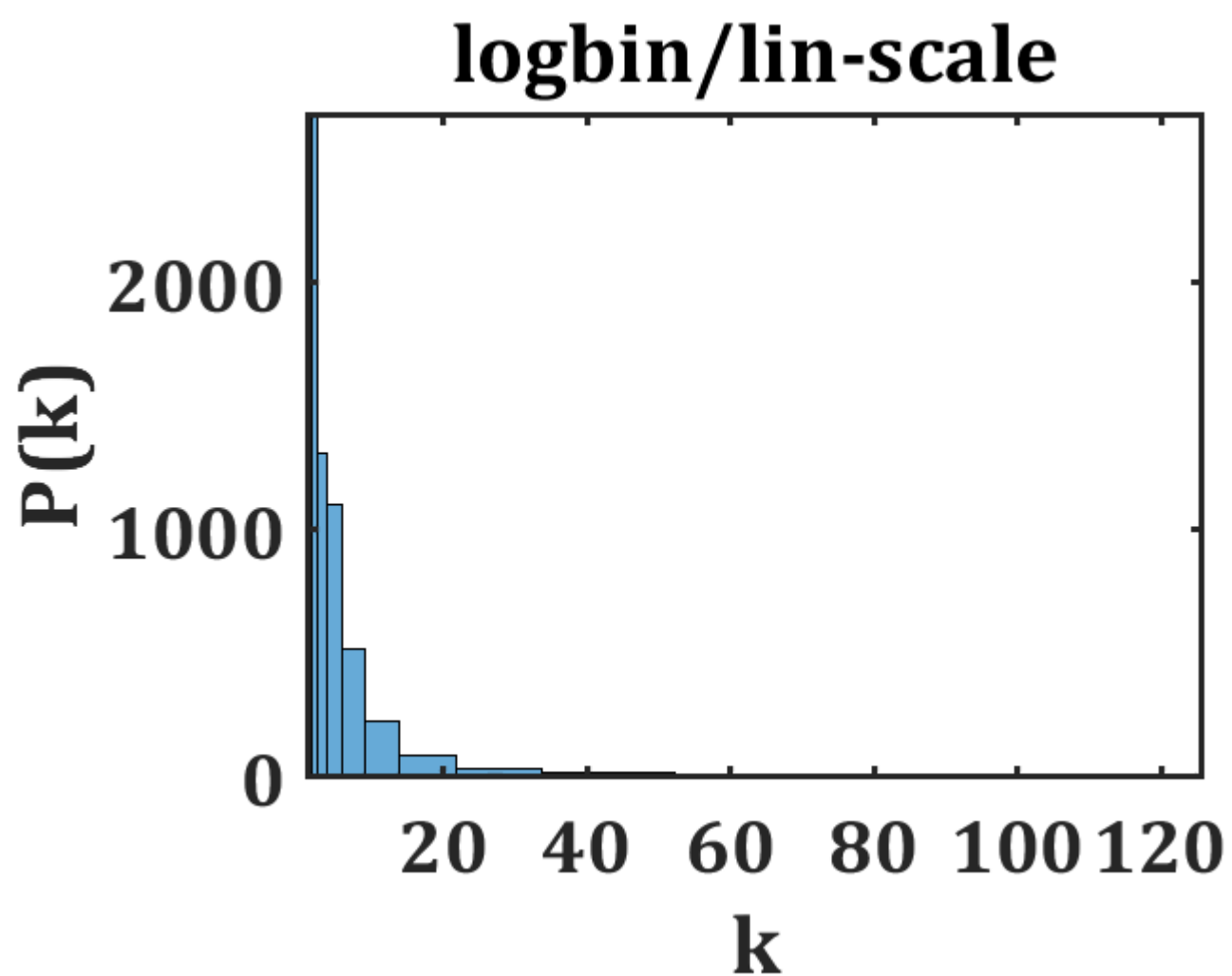
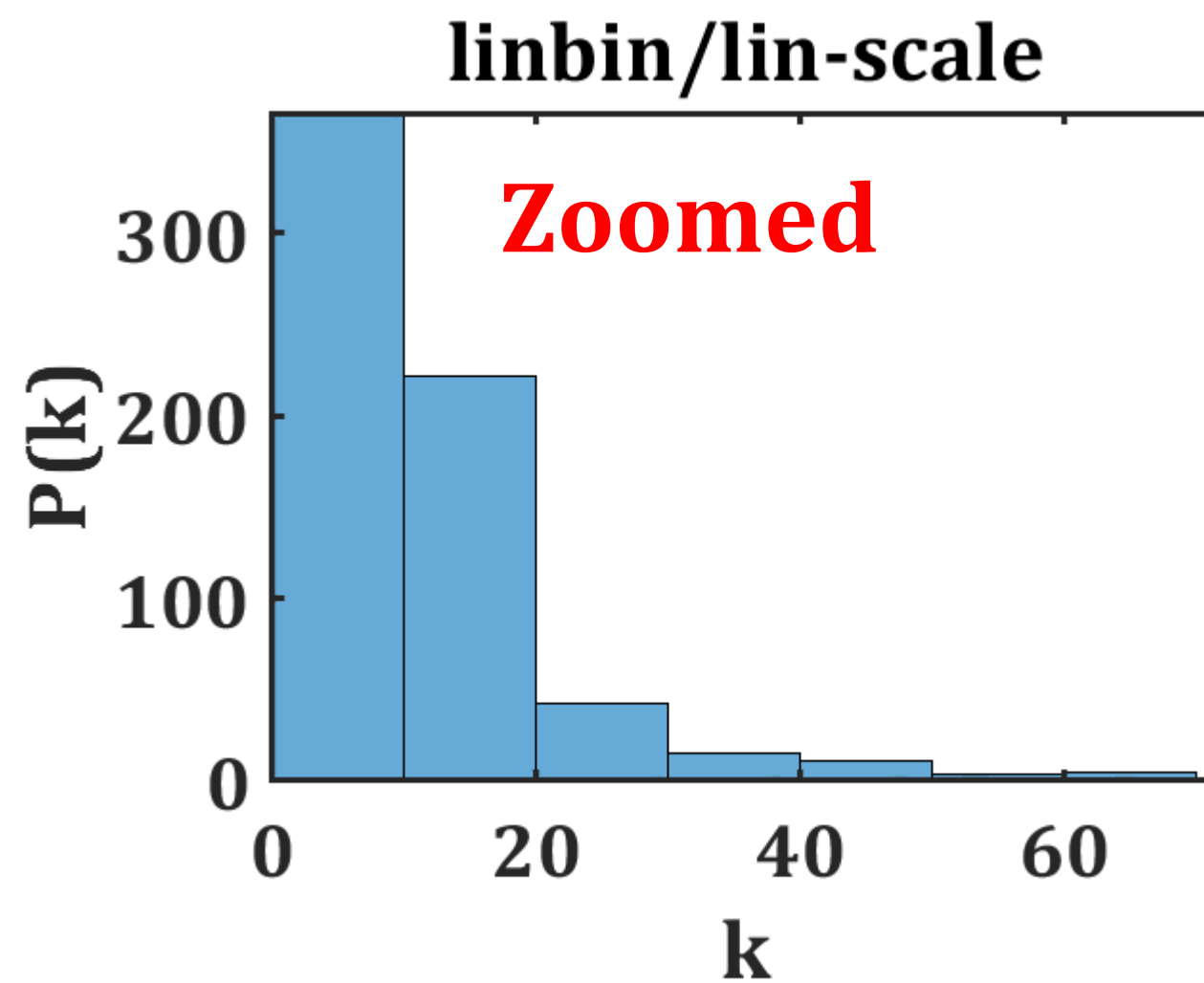
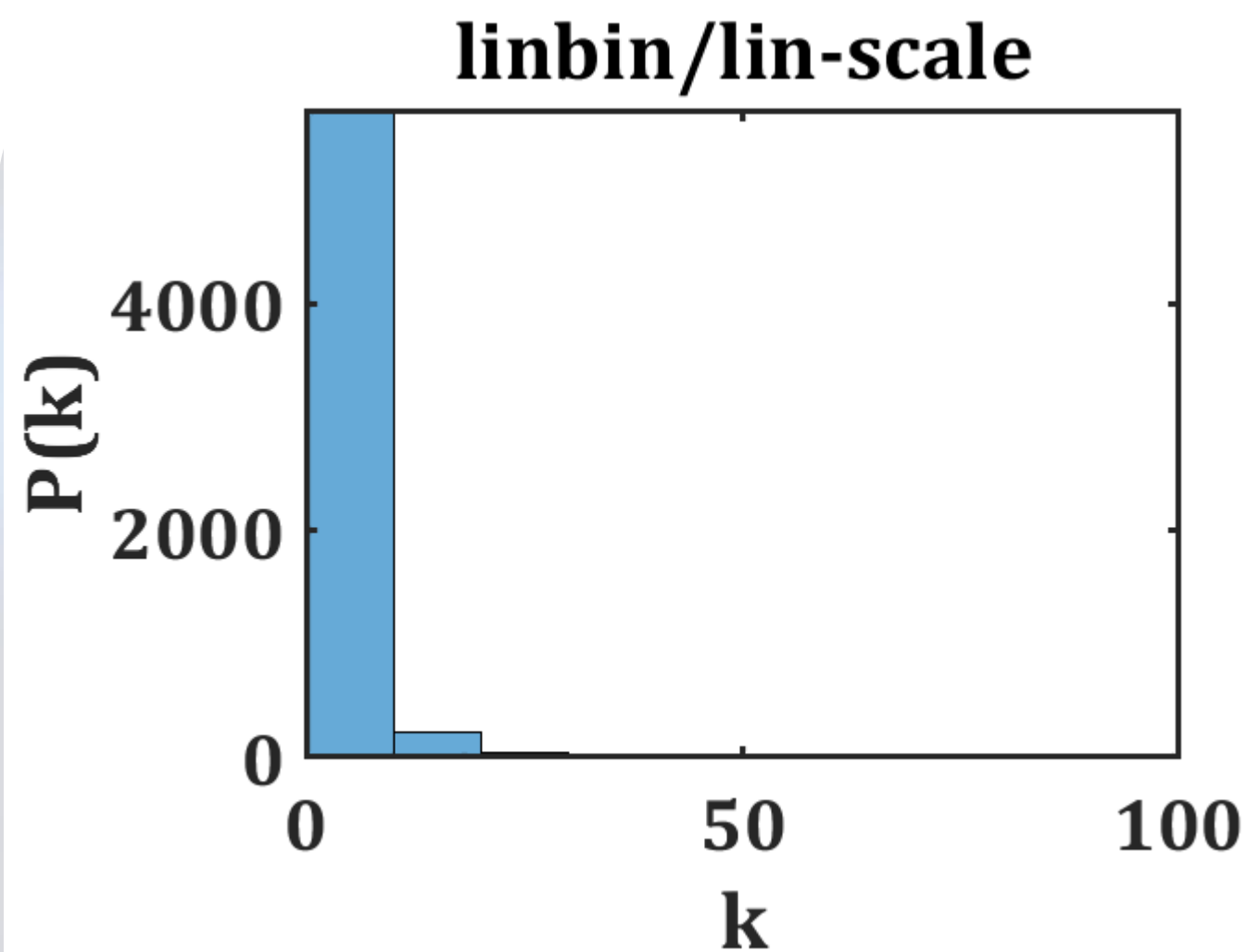
We can also calculate higher moments of the distribution $p(x)$. For instance, the second moment, the mean square, is given by

$$\langle x^2 \rangle = \frac{C}{3 - \alpha} \left[x^{-\alpha+3} \right]_{x_{\min}}^{\infty}.$$

If $\alpha > 3$, then the second moment is finite and well-defined, taking the value

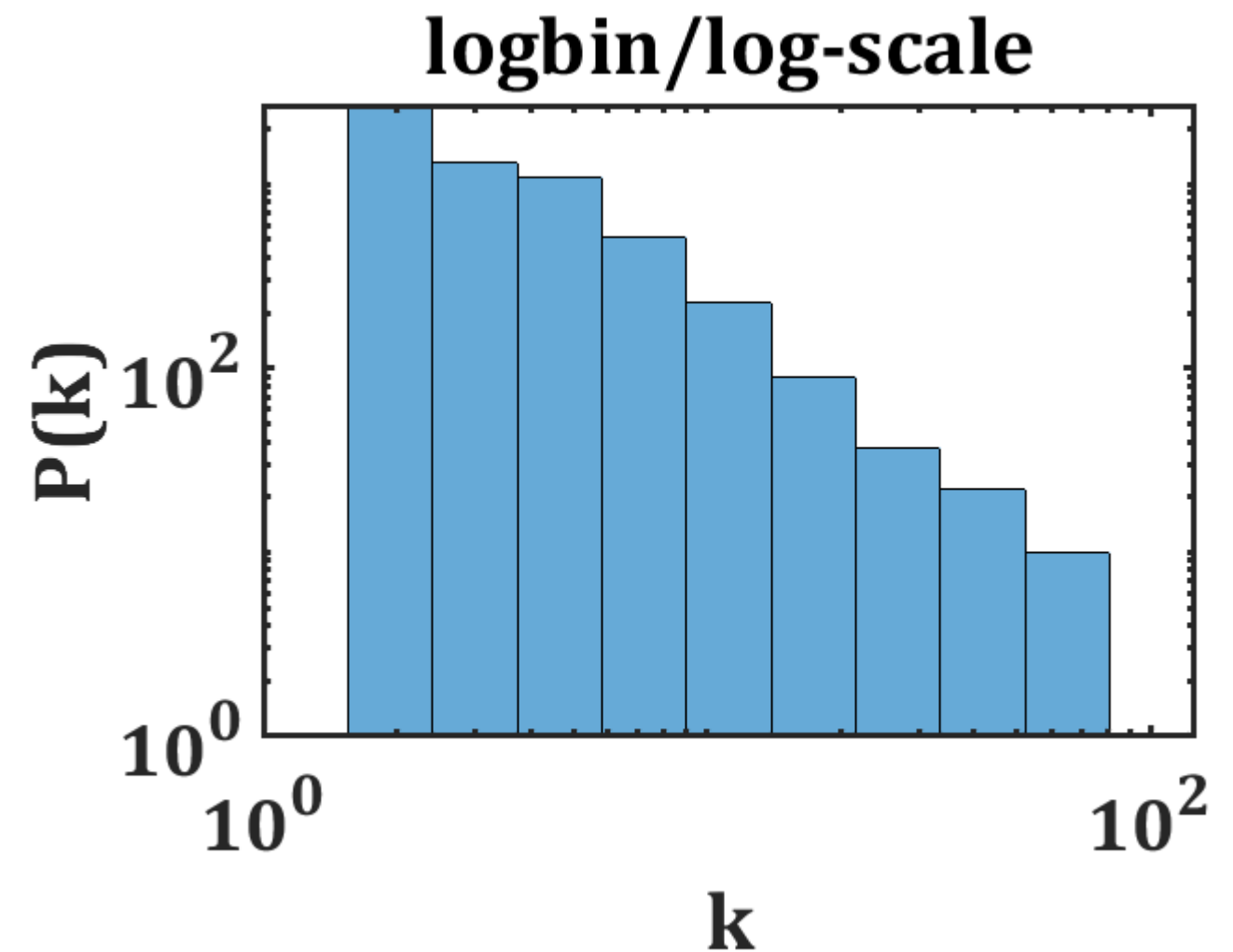
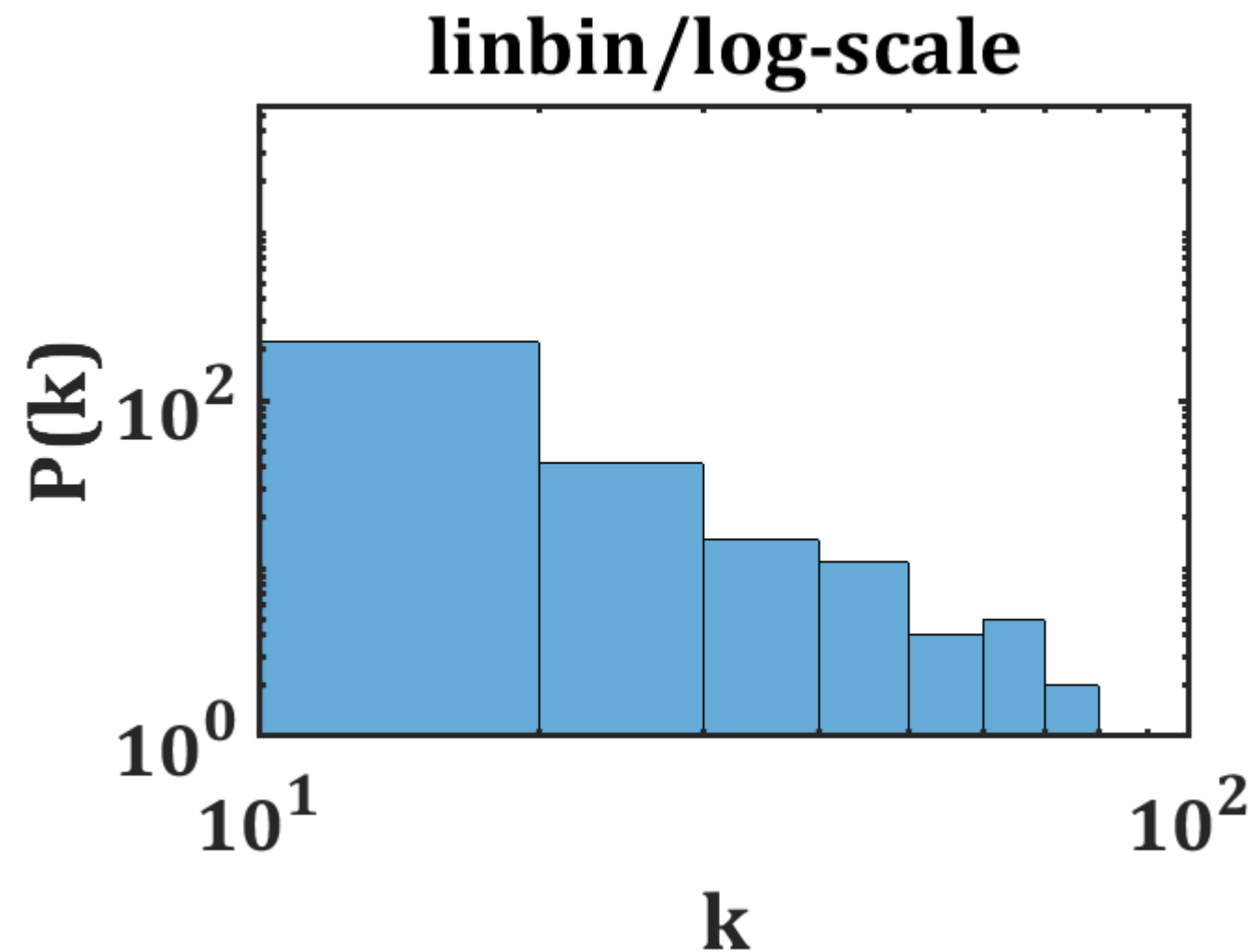
$$\langle x^2 \rangle = \frac{\alpha - 1}{\alpha - 3} x_{\min}^2.$$





Degree Distribution

Log/lin-binning , log scale



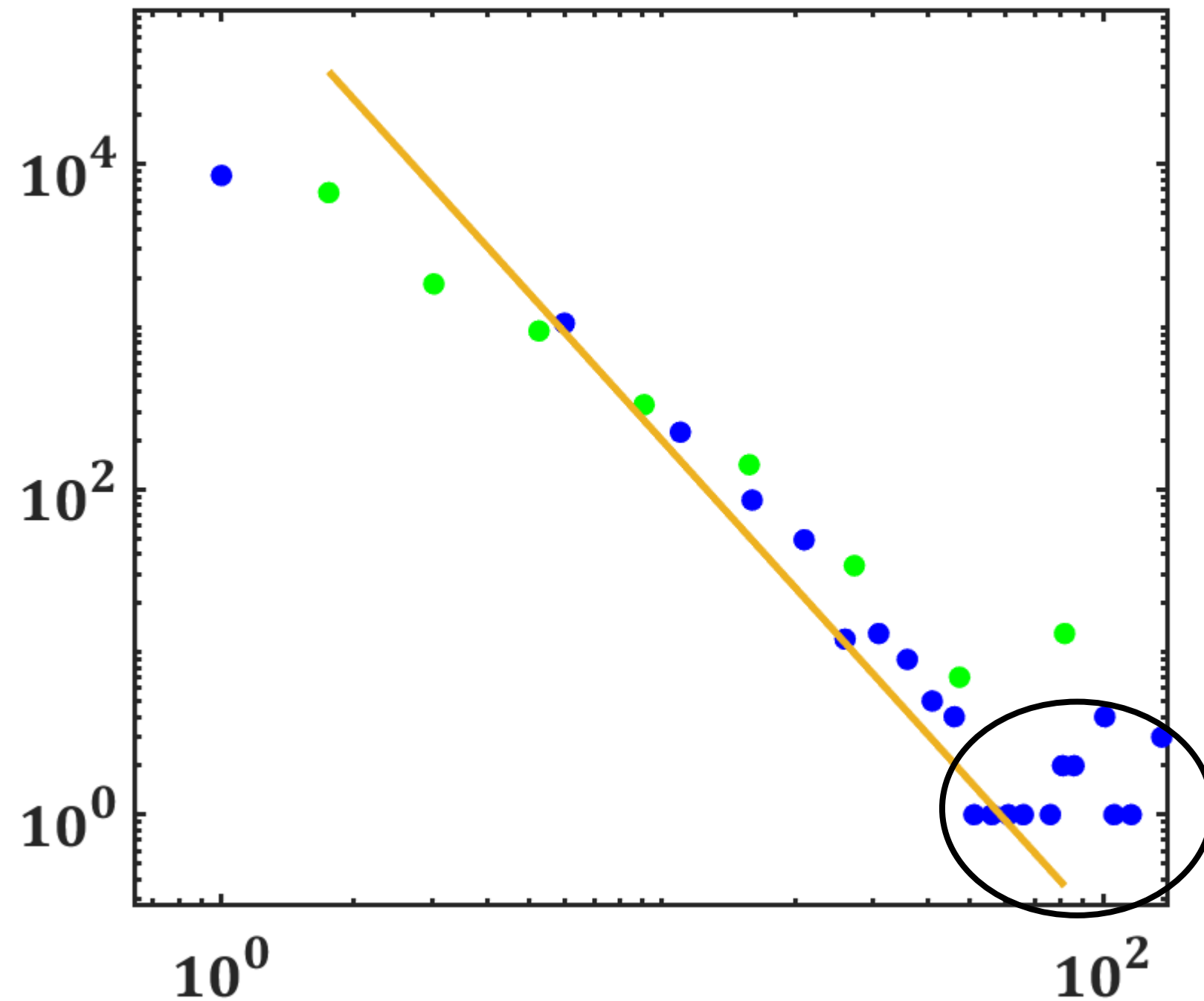
Degree Distribution

Matlab Script

```
load A_7k.mat;
deg=sum(A);
edges = 0.0:10:106;
edges2 = 3.^(0:0.4:4.4);
% %edges=10.^[-1:1:3];
figure(1); histogram(deg,edges);
axis tight;%%% Linear
set(findall(gcf,'-property','FontSize'),'FontName','Cambria',...
'FontSize',24,'linewidth',2.0,'fontweight','b') ;
set(gca, "Xscale", "log")
set(gca, "Yscale", "log")
xlabel('k');
ylabel('P(k)');
title('linbin/log-scale')
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
figure(2); histogram(deg,edges2);
axis tight;
set(findall(gcf,'-property','FontSize'),'FontName','Cambria',...
'FontSize',24,'linewidth',2.0,'fontweight','b')
set(gca, "Xscale", "log");
set(gca, "Yscale", "log");
xlabel('k');
ylabel('P(k)');
title('logbin/log-scale')
```

Degree Distribution (Plotting is tricky!!)

10k node



Degree Distribution (log/lin binning in log-log scale)

```
load('A_6k.mat');
% A=BA(9000,4);
%load A_7k.mat;
deg=sum(A);
ind=find(deg==1);
deg(ind)=2;
edges = 1.1:5:150;
edges2 = 3.^(0.01:0.5:4.7);
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
[Lob, Lob2] = histcounts(deg, edges2);
ds = diff(edges2);
sl = (edges2(1:end-1)+edges2(2:end))*0.5;
nsl = Lob(1:end)./(sum(Lob)*ds);
loglog(sl,nsl,'g.','MarkerSize',30); hold on;
%%%%%%%% Linear bin %%%%%%%%%%%%%%
[LiNb,LiNb1] = histcounts(deg,edges);
ds1 = diff(edges);
slin = (edges(1:end-1)+edges(2:end))*0.5;
loglog(slin,LiNb./(sum(LiNb)*ds1),'b.','MarkerSize',30);

hold on;
% loglog(Lob2(2:end-1),Lob2(2:end-1).^-3,'-','linewidth',3);
loglog(sl,8*sl.^-3,'-','linewidth',3);

set(gcf,'InvertHardCopy','off','Color','white');
set(gcf, 'PaperPositionMode', 'auto','position', [0, 0, 700, 700]);
set(legend,'color','none');
set(legend, 'Box', 'on');
xlabel('$k$', 'Interpreter','LaTeX','FontSize',30);
ylabel('$P(k)$', 'Interpreter','LaTeX','FontSize',30);
xlim([0.8 200]);
ylim([10^-6 10]);
set(findall(gcf,'-property','FontSize'),'FontName','Cambria',...
    'FontSize',24,'linewidth',2.0,'fontweight','b')
```

6k node

Degree Distribution

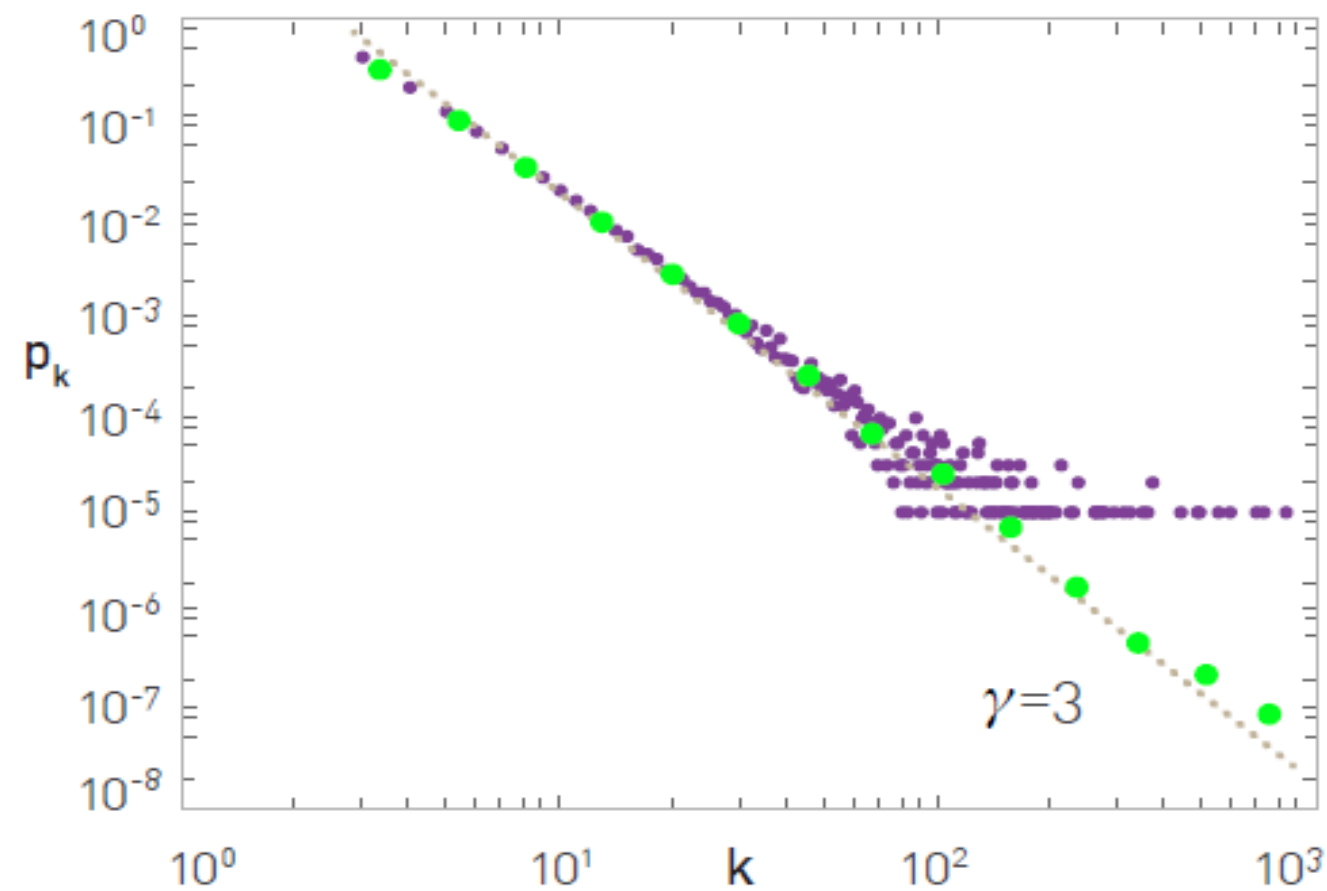


Figure 5.4
The Degree Distribution

The degree distribution of a network generated by the Barabási-Albert model. The figure shows p_k for a single network of size $N=100,000$ and $m=3$. It shows both the linearly-binned (purple) and the log-binned version (green) of p_k . The straight line is added to guide the eye and has slope $\gamma=3$, corresponding to the network's predicted degree exponent.